The International Journal of Life Cycle Assessment (2025) 30:3560–3584
https://doi.org/10.1007/s11367-025-02508-w

SOCIETAL LCA

Towards AI-augmented sustainability assessments: integrating large language models in the case of product social life cycle assessment

Carolyn Cole[1,2] · Arash Hajikhani[1,2] · Eveliina Hylkilä[1] · Essi Paronen[1] · Hanna Pihkola[1]

Received: 30 September 2024 / Accepted: 25 June 2025 / Published online: 8 August 2025
© The Author(s) 2025

Abstract

Purpose As the importance of social responsibility is increasingly recognized, demand for robust social life cycle assessment (S-LCA) has increased. This study presents a novel approach leveraging artificial intelligence (AI) to augment expert-led processes and enhance the efficiency, scalability, and accuracy of S-LCA.

Methods The method utilizes advanced natural language processing (NLP) capabilities and large language models (LLMs) to partially automate the evaluation of social factors such as community engagement, labor practices, and human rights considerations against S-LCA pre-defined criteria. The method is applied in parallel to a standard manual practice of a reference-scale S-LCA on a carton product case study in Finland. Results obtained from the AI-augmented assessments are then compared with those derived from the manual method.

Results and discussion The comparative analysis reveals a 50% agreement rate between the AI and manual assessment outcomes. We find that three driving factors explain the differences in the remaining outcomes. First, outcomes differed where the human evaluators drew on tacit knowledge unavailable to the AI. Second, the human evaluators inherently weigh negative evidence more heavily than positive evidence. And third, outcomes differed in cases where the assessment of a topic was highly sensitive to stakeholder perspective, and the human and AI evaluators assumed differing perspectives in their assessments. Depending on the factor of difference, in some cases, the AI provided a more objective and fair assessment than the human evaluator, while in others, the human evaluator provided a more contextualized and nuanced assessment than the AI.

Conclusions This study contributes to the emerging field of AI-supported assessments by presenting a practical framework for integrating LLMs into S-LCA. The findings aim to inform stakeholders, researchers, and policymakers about the potential benefits and limitations of incorporating AI in evaluation processes that traditionally entail a degree of subjective judgement. The insights gained from the comparative analysis provide valuable considerations for the ongoing development and adoption of AI-assisted approaches in S-LCA and similar evaluation contexts.

Recommendations For future applications of automation in S-LCAs and similar evaluation contexts, we suggest mitigations to minimize differing factors, including priming the AI pipeline with contextual materials and explicitly defining desired nuances in system and task instructions.

Keywords Social life cycle assessment · Social sustainability · Social responsibility · Artificial intelligence · Large language models · Retrieval-augmented generation · Evaluation methods · Automation

1 Introduction

As recognition of the importance of social responsibility has increased in recent years, demand for robust methods to assess the social impacts of products and services has risen, leading to the development of social life cycle assessment (S-LCA) (Chhipi-Shrestha et al. 2015; Arvidsson et al. 2018; Tragnone et al. 2023). S-LCA has the potential to be a powerful tool in the field of sustainable development, as it provides a structured approach to evaluate positive and negative

Communicated by Daniela Sica.

✉ Carolyn Cole
carolyn.cole@vtt.fi

[1] VTT Technical Research Centre of Finland, Espoo, Tekniikantie 21, FI-02150, Finland

[2] School of Business and Management, LUT University, Lappeenranta, Yliopistonkatu, 34, 53850, Finland

social impacts associated with a product or service across its entire life cycle (Ramos Huarachi et al. 2020; UNEP 2020). Yet, despite this potential, the application of S-LCA is not widespread (Tragnone et al. 2023). The reliability and accuracy of data in S-LCA, much like in broader LCA contexts, are hampered by the scarcity of current and comprehensive data, which limits the strategic utility of the assessments (Chhipi-Shrestha et al. 2015; Arzoumanidis et al. 2017; Sassanelli et al. 2019). Adoption of the methodology is hindered by challenges in data collection and processing, as well as a large variety of applied methodological approaches. In response to these challenges, scholars have called for the integration of digital tools and data science applications to support the S-LCA process (Tokede and Traverso 2020; Sica et al. 2022).

Recent leaps in the field of natural language processing (NLP), particularly in the development of large language models (LLMs), mark a unique opportunity for addressing some of the limitations in current S-LCA practice. The field of NLP has been profoundly transformed by the rapid advancement of LLMs. These models, such as OpenAI's Generative Pretrained Transformer (GPT) series, have introduced unprecedented capabilities in text summarization, classification, and understanding (Devlin et al. 2018; Brown et al. 2020; Raffel et al. 2020; Yoo et al. 2021; Min et al. 2021). Building on these capabilities, frameworks enabling the integration of LLMs with external knowledge sources have emerged, allowing users to develop highly complex yet competent NLP pipelines for knowledge-intensive tasks (Lewis et al. 2020; Pokhrel et al. 2024).

LLMs have been successfully integrated into qualitative data collection and analysis workflows (Fisher & Biemann, 2024), including the extraction of ESG data for corporate social responsibility research (Bronzini et al. 2024; Anagnostopoulos et al. 2025), metadata collection from scientific literature (Polak and Morgan 2024; Song et al. 2025), and contextualized content extraction in media analysis (Törnberg 2024). The collection of S-LCA data requires contextualized interpretation of social issues and comprehension of nuance, which have traditionally been considered distinctly human capabilities. However, recent research suggests that contemporary LLMs are also capable of contextualized interpretation, even outperforming human annotators in some cases (Törnberg 2024; Polak and Morgan 2024). Research applying LLMs for social data extraction is nascent, however, and to our knowledge, the present study represents the first augmentation of S-LCA data collection and social performance assessment.

This study explores and develops a practical application of LLMs in S-LCA to support data collection and processing as part of a qualitative reference-scale assessment. Incorporating LLMs into the S-LCA framework produced out of the EU Horizon's ORIENTING project offers a novel approach to overcoming the traditional challenges of the S-LCA methodology (for the ORIENTING project report detailing the S-LCA framework, see Zanchi et al. (2023)). The ORIENTING S-LCA performance assessment template provides a clear framework for evaluating data against explicit social impact criteria. By automating the extraction and analysis of qualitative data, the AI-augmented method developed in this study aims to significantly reduce the time and effort involved in manual evaluations while improving the consistency and objectivity of the assessments. This could help address some of the current bottlenecks faced, especially in the context of reference-scale assessments. Through a comparative analysis with a fully manual application of the method, we further explore the nuances of S-LCA and AI integration in this process. Empirical data for testing this approach was derived from an S-LCA case study, which was conducted for a beverage carton manufactured in Sweden (for the full manual S-LCA case study, see Pihkola et al. (2024)). For the purposes of testing the AI-augmented method, 10 social topics were selected from the full set of social topics defined for the reference scale developed in the ORIENTING project.

We find that manual S-LCA evaluations may incorporate background knowledge relevant to the assessment but unavailable in the data explicitly available (e.g., country context). However, the AI evaluation results in assessments consistent with a given S-LCA criteria across implementations, and therefore offers more objective evaluations. The results of the study offer insights into the effectiveness of AI-augmented evaluations in S-LCA and identify areas for further refinement. Human judgment is needed to make sure that case and context-specific nuances are taken into account and to confirm that the assessment is sensible.

The paper will continue with a short background describing the challenges faced in S-LCA implementation and adoption, and the suitability of LLM applications to these particular problems. Then, the ORIENTING S-LCA framework and its manual application are explained before a detailed walk-through of the AI augmentation. The outcomes of each application are presented, and an analytical comparison is conducted to explore driving factors for any differences found in the results. The paper concludes with a discussion of these driving factors and recommendations for future applications of AI in S-LCA and similar evaluation contexts.

2 Background

2.1 Challenges to the implementation and adoption of S-LCA

S-LCA is a methodology developed to assess the social impacts of products and services, accounting for their whole

life cycles. The methodology is based on interlinked stakeholders and social sustainability topics for which potential impacts are evaluated (UNEP 2020). S-LCA has been characterized by Ramos Huarachi et al. (2020) as a crucial tool in the field of sustainable development. Yet, at the same time, Pollok et al. (2021) consider it an immature method. Despite the importance of S-LCA for the advancement of social impact assessment in practice, the method is vulnerable to limitations which are preventing its wider application. The main challenges are found in data collection and processing, and variability in assessment.

S-LCA is conducted based on data collected across the value chain, but the data collection process is considered time-consuming and complex (Tokede and Traverso 2020; UNEP 2020; Pollok et al. 2021). The issue is further exacerbated in cases where the data is simply not accessible (Chhipi-Shrestha et al. 2015; Tokede and Traverso 2020; Huertas-Valdivia et al. 2020). This presents an immediate barrier, as S-LCA requires access to data which is consistent and representative (UNEP 2020). Similar challenges are noted in traditional LCA practices, where the Life Cycle Inventory (LCI) phase often suffers from insufficient data availability, leading to a reliance on estimates or the exclusion of certain stages in the life cycle (Notarnicola et al. 2017; Arzoumanidis et al. 2017). In cases where the data is of poor quality or cannot be collected, unit processes may similarly need to be excluded in S-LCA, which can significantly impact both the coverage and the reliability of the S-LCA results. If primary data is not available, secondary data, often compiled for other purposes, must be used, which may also introduce compromising factors to the S-LCA results (UNEP 2020). Methodological comparability issues among different databases, and the generic nature of the database data, contribute to further variability and uncertainty in S-LCA outcomes, making it difficult for managers to utilize these insights in decision-making (Cooper et al. 2013; Zhang et al. 2020).

Currently, the United Nations Environmental Programme (UNEP) S-LCA guidelines (2020) propose prioritization as a way to improve the data collection process. According to UNEP (2021), the recommended actions for prioritization include carrying out a literature review to uncover key social issues, to define the most intensive activities in the studied system, and to identify hotspots, taking into account the life cycle of the product. Currently, leveraging digital technologies such as blockchain, IoT, and big data analysis has been advocated to enhance the accuracy and reliability of data in LCA studies (Nascimento et al. 2019; Bai et al. 2020), suggesting potential applicability in enhancing S-LCA methodologies, as well. Tokede and Traverso (2020) propose the use of machine learning and data science to facilitate data collection in S-LCA. The authors suggest that with these technical solutions, the efficiency of the data collection can be refined, though only in an assisting role. Similarly, Ertz et al. (2024) synthesize the relevant literature and suggest that digitalization, such as AI, in the S-LCA domain should be researched and explored.

In addition to data challenges, a second identified deficit in S-LCA methodology is variability in methodological choices. The existence of different methods for the impact assessment step, namely the impact-pathway approach and the reference-scale approach, makes comparisons between different S-LCA studies very difficult. While some of the identified challenges concern the S-LCA method as a whole, different digital and AI-based solutions are most likely needed to address specific challenges related to the impact assessment step conducted using the impact pathway and the reference-scale assessment. Often, individual choices of the researcher form the base for the S-LCA (Huertas-Valdivia et al. 2020). Pollok et al. (2021) conclude that S-LCA is missing transparency at key methodological junctures, such as the selection of indicators and interpretation of data. The issue of subjectivity can affect S-LCA in the data collection phase (UNEP 2020) and in scoring the social impacts (Huertas-Valdivia et al. 2020), particularly in the case of reference-scale assessment, which is the focus of this paper. While a lack of transparency related to methodological choices can be addressed with improved reporting practices, augmenting S-LCA with AI could be helpful for the practitioners by indicating if proposed scores in the reference-scale assessment are not supported by the evidence found. On the other hand, the need to collect and handle large amounts of social data (covering different social topics and stakeholder groups) from the product life cycle is a shared challenge for all S-LCA applications. AI-assisted solutions that support the collection and analysis of social data from various sources could be helpful, despite the specific methodological approach or impact assessment method selected for S-LCA.

2.2 Application of LLMs in knowledge-intensive tasks

LLMs are a subset of AI that excel in a variety of natural language tasks including question answering, translation, and sentiment analysis, enabling them to process large quantities of text with a speed and accuracy surpassing human capabilities (Wang et al. 2018; Jiang et al. 2024). These AI systems have substantially advanced the automation of knowledge-intensive tasks that traditionally require significant human effort and expertise. For instance, frameworks like LangChain, an open-source tool, allow users to leverage LLMs to create complex NLP pipelines. This framework combines LLMs with external knowledge sources such as databases and application programming interfaces (APIs), facilitating the development

of applications that require deep language understanding (Topsakal and Akinci 2023; Pokhrel et al. 2024). At the time of this study, the LLMs exhibiting the highest performance in knowledge-intensive tasks were OpenAI's GPT 3.5 and recently released GPT 4 models (Artificial Analysis 2025), which this study employed. Table 6, listing the high-performing models at the time of this study, can be found in the appendix.

One innovative approach in this field is Retrieval-Augmented Generation (RAG), which improves LLM outputs by integrating external knowledge retrieval. This technique first retrieves relevant information using a search engine, then incorporates this data into the LLM processing, enabling the model to provide more detailed and context-specific answers (Lewis et al. 2020; Gao et al. 2023). The utility of LLMs and tools like LangChain extends across various domains. These models have been successfully applied in automating the analysis of legal documents, aiding medical diagnosis, and creating educational content (Ziegler et al. 2019; Chalkidis et al. 2020; Liu et al. 2021). Such applications streamline complex tasks and enhance the accuracy and consistency of the outcomes.

Rather than replacing expert judgment, LLMs are increasingly used to augment it, scaling initial data collection while supporting human-in-the-loop interpretive processes (Fischer and Biemann 2024). In the context of social data collection, LLMs have been successfully applied in cases of ESG data extraction from sustainability documentation and nuanced annotation of scientific literature and social media text, demonstrating higher consistency and efficiency than traditional data extraction and qualitative coding approaches in some cases (Törnberg 2024; Polak and Morgan 2024; Anagnostopoulos et al. 2025). However, LLMs' use in social data workflows is not without limitations. As Rask and Shimizu (2024) highlight, reliance on LLMs for synthetic data generation introduces potential biases inherited from the models' training data, which could shape the perspective of generated content in tasks that require judgement or opinion. While the authors consider this specifically in the context of synthetic survey data generation, the issue is also relevant to the social assessment stage of S-LCA, where the evaluation of social data against a social topic may vary depending on the stakeholder perspective taken.

Research on the development and deployment of AI personas to represent and generate content from specified perspectives is emerging and shows promise in addressing these issues (Arora et al. 2024; Sun et al. 2024). Future work on the integration of LLMs for S-LCA could benefit from learnings in this research stream as the scholarship develops. However, for the purposes of this study, an exploration of AI personas is out of scope.

2.3 Technological integration for S-LCA

Ongoing advancements in digital technologies, as evidenced in LCA studies, support the integration of similar technologies in S-LCA to overcome data collection and analysis challenges, thereby potentially enhancing the effectiveness and reliability of S-LCA results (Wang et al. 2019). LLMs offer potential for significant advancements. S-LCA using reference-scale assessment relies on extensive text-based data to evaluate social impacts across multiple stakeholders, organizations, and life cycle stages. The process involves the collection and analysis of data from diverse sources, including company reports and stakeholder interviews (UNEP 2020). This has been a manual and time-consuming task, prone to subjectivity and inconsistency.

The role of digital technologies in supporting S-LCA studies is still at an early stage in the literature, with a need for more operational research and analysis on specific S-LCA phases (Sica et al. 2022). By integrating LLMs with frameworks like LangChain and techniques such as RAG, the efficiency and robustness of S-LCA processes can be improved. LLMs can facilitate the automatic extraction of relevant information from a variety of unstructured data sources and align it with specific S-LCA criteria. This not only reduces the manual effort required but also ensures a more comprehensive and consistent evaluation of social impacts. This consistency is important as traditional LCA analyses often suffer from strong results variability due to methodological limitations and data uncertainties (Cooper et al. 2013; Arzoumanidis et al. 2017). Additionally, the capability of LLMs to generate human-like explanations allows for greater transparency and interpretability of assessments, fostering better communication with stakeholders and decision-makers. In combination with a robust evaluation framework, it becomes possible to verify the reliability and validity of the inferences made by LLMs, ensuring they are contextually grounded and relevant to the posed questions (Gao et al. 2023; TruEra 2024a). Integration of LLMs can advance S-LCA by making it more efficient and replicable, and providing better possibilities for increasing the share of primary data collected from the value chain.

3 Methods

3.1 Manual S-LCA

The ORIENTING S-LCA methodology covers 29 social topics (Zanchi et al. 2023). The S-LCA methodology developed in ORIENTING takes foundations from the UNEP Guidelines for S-LCA (UNEP 2020) and the Handbook for Product Social Impact Assessment (Goedkoop et al. 2020).

An interim version of the ORIENTING S-LCA methodology with 28 social topics was tested in a case study, which was conducted for a carton product manufactured in Sweden and Finland with a global upstream value chain (see Pihkola et al. (2024) for details of the case study). Within the original case study, 23 social topics were evaluated. Selection of topics was done based on a materiality assessment, which was conducted together with the case company. From the original S-LCA case study, collected data and reference-scale assessment results for 10 social topics were utilized in this paper to provide a proof-of-concept for AI-augmented S-LCA. For the selected 10 social topics, both manual assessment and an LLM-assisted evaluation were conducted. The 10 social topics were selected by researchers who had been involved in the manual assessment and were chosen to represent diversity in topic nuance, ease (and difficulty) of agreement between human reviewers on the topic's assessment, and public availability of the data cited as evidence in the manual assessment.

At the core of the applied S-LCA methodology is a five-level reference scale, which was developed during the ORIENTING project. Using the pre-defined reference-scale assessment template, social performances of actors participating in different life cycle stages are scored to the different social topics, ranging from positive (+2) to negative (−2) performance (Zanchi et al. 2023).[1]

Data sources for the reference-scale assessment can include both internal materials (such as code of conduct documents and internal guidelines) and external documents (published policies, statements, statistics, and annual reports). Supplier websites, news, and media can be used as additional information. Relevant sources shall be cited for each social topic and related assessment score. Within the ORIENTING S-LCA methodology, the recommendation is to apply a combination of social performance assessment (based on a reference-scale assessment) using primary data collected from processes managed by the company, and social risk assessment based on secondary data (e.g., from social databases) for those parts of the value chain for which primary data is not yet available. According to the ORIENTING S-LCA methodology, eventually, the aim is to increase the share of performance assessment and primary data coverage. However, the wide-scale application of social performance assessment is currently challenged by the time-consuming manual data collection and analysis phase.

[1] The performance assessment template (spreadsheet), which includes definitions for all the social topics included in the ORIENTING S-LCA methodology, together with the pre-defined reference scale, which was used in our case study, can be downloaded from the appendix included in Zanchi et al. (2023), which is available for download from: https://ec.europa.eu/research/participants/documents/downloadPublic?documentIds=080166e507e98507&appId=PPGMS

For a full description of the original S-LCA case study, refer to Pihkola et al. (2024). For a full description of the ORIENTING S-LCA method, refer to Zanchi et al (2023). For the purposes of this study, the definitions of the 10 social topics tested and the criteria for their reference scale scoring are provided in an appendix.

3.2 AI-augmented S-LCA

The AI-augmented S-LCA process developed in this study aims to replicate the manual assessment using computational means in place of a human agent. To achieve this, the AI-augmented process is separated into four stages, allowing for evaluation of the parameters and outcomes at key junctures. Figure 1 displays a high-level overview of these AI-augmented stages. The first stage and second stage are parallel, consisting of data preparation and query design, respectively. The third stage extracts social topic information from each data source, and the fourth stage uses that information to determine the level that the organization falls under for the social topic in question. The third and fourth stages employ Retrieval-Augmented Generation (RAG) to enhance the accuracy and credibility of the process, and evaluation metrics of the RAG implementations are generated to determine the validity and reliability of the outputs. In S-LCA terminology, the three stages would belong to the social inventory phase, whereas the fourth phase would be part of the social impact assessment and interpretation phases. Here, the definitions of social topics, indicators, and reference scales as defined for the ORIENTING S-LCA methodology were applied as necessary building blocks for the LLM in the AI-augmented framework. If such pre-defined social topic definitions and reference scales are not available, they would first need to be constructed, increasing the required workload.

The next section describes the RAG implementation and evaluation framework. Then, the stages of automation in the S-LCA are discussed in more detail.

3.2.1 Retrieval-Augmented Generation: application and evaluation

RAG has been shown to enhance the deployment of LLMs for knowledge-intensive tasks by integrating external data with the LLM's intrinsic knowledge (Lewis et al. 2020; Gao et al. 2023). The RAG process converts user prompts into semantic data, which is used to retrieve information from external sources. The retrieved data is then integrated with the LLM's internal knowledge to generate a more accurate and trustworthy response. RAG offers several benefits, including the ability to provide responses based on current or curated information, increase user trust by allowing source traceback, reduce hallucinations, and lower computational costs. Frameworks have emerged to evaluate

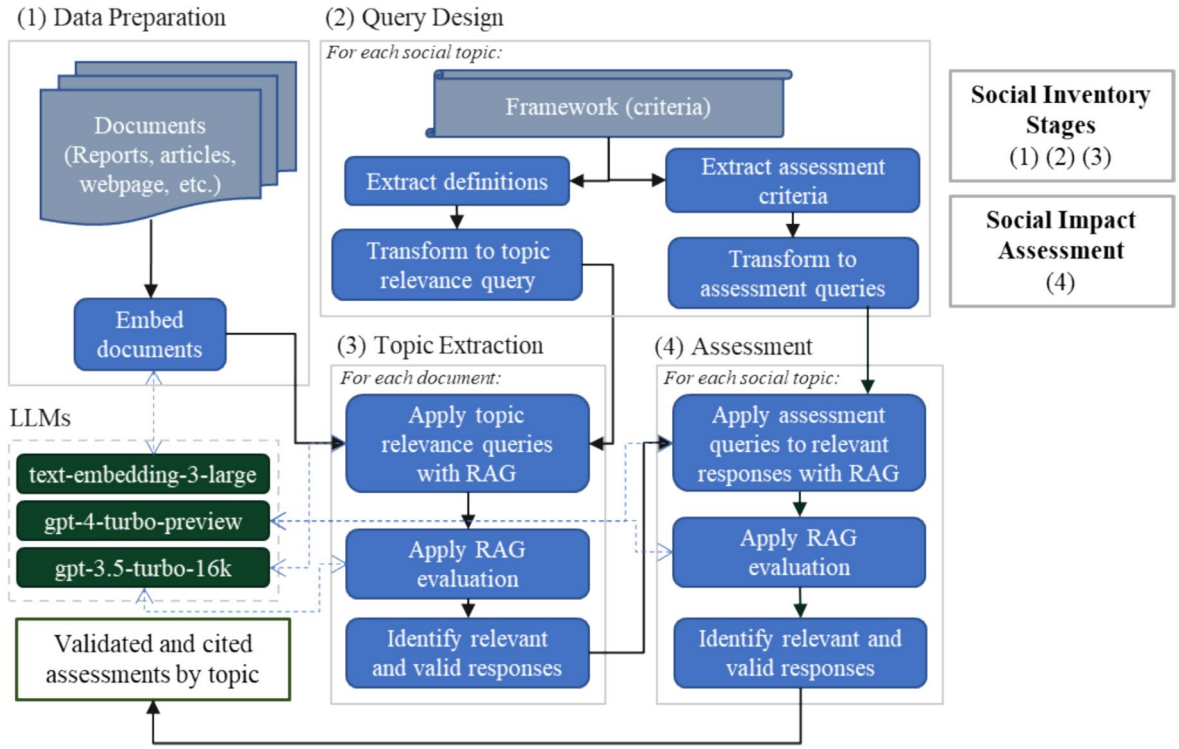


Fig. 1 Process chart for LLM-driven assessment pipeline

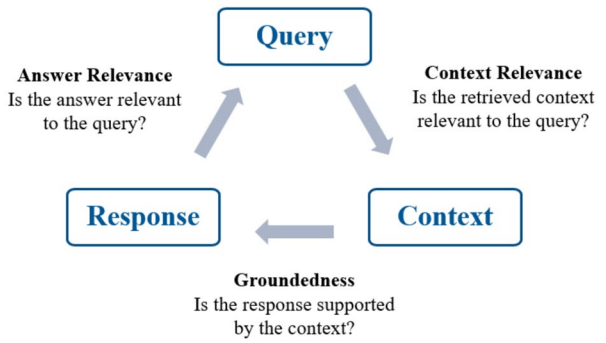


Fig. 2 The TruLens RAG Triad (TruEra 2024a)

the performance of RAG instances, including the TruLens RAG Triad, which this study employs (TruEra 2024a). The TruLens RAG Triad targets two main aspects, Retrieval Quality and Generation Quality, through evaluations of the dimensions, Context Relevance, Groundedness (sometimes referred to in the literature as Faithfulness), and Answer Relevance, as shown in Fig. 2. A RAG instance and the Trulens RAG Triad are applied in stages three (Topic Extraction) and four (Assessment) of this AI-augmented S-LCA process.

Context Relevance compares the query to the material it is referencing to provide an answer. In the Topic Extraction stage, this indicates whether a given document is relevant to a social topic. At the Assessment stage, this indicates whether the information retrieved from documents relevant to a social topic is further relevant to any of the level criteria defined for that social topic. Groundedness compares the result of the query to the material it referenced to generate the response. This indicates whether there is evidence in the material to support the response. Finally, Answer Relevance compares the result of the query to the query itself. This indicates how relevant the model's response is to the query supplied.

We utilized TruEra's open-source Python library, TruLens, to implement the RAG Triad evaluations. Specifically, we applied the TruLens feedback functions "groundedness," "context relevance," and "relevance" (the function representing Answer Relevance), each of which produces an evaluation metric between 0 and 1. These functions operate by applying a scoring framework specifically tailored to the evaluation dimension in question, then computing an average over these scores and normalizing it to a 0 to 1 scale. For instance, the Groundedness metric involves parsing the LLM response into individual sentences and comparing each sentence with the context retrieved from the data source used to answer the query. Each comparison is scored on a scale from 1 to 10, and the normalized average of these ratings constitutes the Groundedness metric. The Context Relevance metric follows a similar procedure but compares the query to the retrieved context. The Answer Relevance metric compares

the query to the LLM responses using the same process. These functions and their scoring frameworks are publicly available in the TruLens GitHub repository (TruEra 2024b).

3.2.2 Stage 1: data preparation

Thirty-five articles, webpages, and PDFs were cited in the manual rendition of this S-LCA. Of these, two citation URLs led to different storage locations for the same source, and one citation could not be retrieved. This resulted in a set of 33 documents compiled as the input for the AI-augmented S-LCA. Table 7, listing the URLs from which the documents were collected, is given in the appendix. The documents consisted of ten company newsroom (press release) articles, six company sustainability disclosure articles, four company guidelines (e.g., human rights, code of conduct, health and safety, supplier terms and conditions), three annual and sustainability reports, one local news article, and nine miscellaneous company webpages (e.g., legal notice, about us page). The text in each document was extracted and transformed into numerical representations for computational processing using OpenAI's *text-embedding-3-large* embedding model in March 2024.

3.2.3 Stage 2: query design

The AI-augmented S-LCA engages a two-step query process. The first query extracts topic relevance and is applied in stage 3 (Topic Extraction). The second query assigns a level for the social topic.

The interim version of the ORIENTING S-LCA methodology included 28 social topics (ORIENTING D2.3 2022). For each social topic, a definition is provided in two parts: first, what the social topic is about, and second, what the framework aims to measure about the social topic. Then, for each of the five levels (−2 to +2), a definition is given with respect to what kind of behavior or evidence qualifies the level, as well as indicators the evaluator can look for that would suggest the given level as a suitable match. A list of the definitions (what the topic is about and what should be measured) and the reference scale indicators for each of the 10 social topics included in this study are provided in an appendix. Further details on the S-LCA methodology and case study are given in Sect. 3.1 and in Pihkola et al. (2024).

The first query uses the "What the social topic is about" and the "What we want to measure" elements of the criteria framework definitions to formulate an information extraction query for each social topic. The second query uses the criteria framework's "level − 2" to "level + 2" definitions and indicators to formulate an assessment query for each social topic. The template to derive the queries for each social topic is shown in Table 1. Query examples are provided in Table 8 in the appendix.

3.2.4 Stage 3: topic extraction

The third stage applies query 1 of each social topic to each document. The output is a text response summarizing information relevant to the social topic from the document. We implement an RAG infrastructure and create a new retriever for each document. A record of each query and response is generated, allowing for information traceback and RAG Triad evaluation. OpenAI's *gpt-3.5-turbo-16 k* model was used for the topic extraction query in this study in March 2024, with the top-p parameter set to 0.75.

In the Topic Extraction stage, an answer is considered "valid" if it meets one of two scenarios: 1) all three RAG Triad metrics are greater than 0.2; or 2) both Context Relevance and Groundedness are less than or equal to 0.2 and Answer Relevance is greater than 0.2. To arrive at this threshold, a sample of results across documents and social topics was manually reviewed to determine a suitable cutoff. This entailed reviewing the LLM response in the context of the query and the input text and determining a threshold at which the LLM typically reflected and represented the information it was provided in an complete and accurate manner. The threshold of 0.2 was selected for the Topic Extraction stage, ensuring cases with no relevant information were excluded while minimizing drop out of cases with relevant information before the Assessment stage. The first case for validity suggests that the document has relevance to the social topic, the answer to the query is evidenced in the

Table 1 General LLM query formats

Query 1 (Stage 3)	{*Definition—What the social topic is about*}. Your task is to summarize, based on the documents, whether and to what extent {*Definition – What we want to measure*}. Avoid using jargon language, be concise and clear, and deliver only information that is retrieved in the text. If available, capture concise details and protocols mentioned in the report, including any examples or cases. Do not exceed 300 words in your response. If there is no discussion or mention of the topic, respond "no information" and do not elaborate
Query 2 (Stage 4)	Level { *x*}: {*level definition*}. Indicators: {*level indicators*} Your task is to evaluate the given text and determine which level it belongs to. Indicate your judgment in quotation regarding the level for example if the given description falls under Level + 1, indicate "Level + 1" and then after ":" give your chain of thought reasoning for your choice based on the given text and the definitions and performance indicators provided above

document, and the answer has relevance to the query. The second case for validity suggests the document has little to no relevance to the social topic, so the answer is unfounded. For instance, if the answer is "no information," Groundedness will return a low metric because this statement on its own (outside of the context of the query) has no relevance to the information in the document. The query is re-run a maximum of three times until either of these scenarios is met.

If neither scenario is met over these attempts, the answer is considered "invalid." An invalid answer is one in which Answer Relevance is low (0.2 or less), suggesting the result is not relevant to the query, or in which the Context Relevance and Groundedness metrics do not align. For instance, if the Context Relevance is high and the Groundedness metric is low, this suggests that the document contains relevant information, but the answer does not reflect that information. If the Groundedness metric is high and the Context Relevance is low, this suggests that the document has low or no relevance to the social topic, and the answer is overemphasizing the information retrieved.

Only "valid" answers with all three metrics greater than or equal to 0.2 are passed to the Assessment stage. These represent results where the answer has been validated and its content is verified as relevant to the given social topic.

At the time of deployment, OpenAI also offered a very early version of *gpt-4*, as *gpt-4-turbo-preview*. Testing and comparison showed that *gpt-3.5-turbo-16 k* had higher validity scores in the evaluation results than *gpt-4-turbo-preview* for this summarization task. When using *gpt-3.5-turbo-16 k*, the validity evaluation resulted in 81.6% valid responses, while *gpt-4-turbo-preview* resulted in 69.7% valid results. As a result, *gpt-3.5-turbo-16 k* was used to perform the Topic Extraction stage.

3.2.5 Stage 4: assessment

Stage four applies query 2 of each social topic to the summarized information retrieved in stage three. The output is a text response assigning a level assessment and explanation of the level choice. We implement an RAG infrastructure and create a single retriever across the output passed from stage three for the social topic. The RAG evaluation with Trulens' RAG Triad is once again applied.

The Assessment stage uses a different LLM than the Topic Extraction stage, and therefore, the validity threshold for the evaluation metrics needed to be independently defined. Responses were manually reviewed in the context of the information fed into the assessment and the query contents, and based on this, a threshold of 0.4 was determined to be most suitable. In the Assessment stage, an answer to the assessment query is considered valid only if the Answer Relevance and Groundedness metrics are greater than or equal to 0.4. The Context Relevance metric is not meaningful in this stage since the query contains the criteria necessary for all five levels. Since the information gathered in the Topic Extraction stage should indicate alignment with only one level, the content will have no or low relevance to the remaining levels described in the query. As a result, the Context Relevance metric will naturally be low.

This stage was tested with OpenAI's *gpt-3.5-turbo* and *gpt-4-turbo-preview* models. Results using *gpt-3.5-turbo* were of poor quality with low validity. In contrast, results using *gpt-4-turbo-preview* were of higher quality, and for each social topic where an assessment was possible (i.e., input was passed through from stage three), responses were "valid." As a result, OpenAI's *gpt-4-turbo-preview* model was used for the assessment query.

This testing exercise highlights the importance of model choice in different use cases, as the quality of results is highly dependent on model choice. While *gpt-3.5-turbo-16 k* was well suited to the information summarization task in stage three, the higher reasoning capabilities of *gpt-4-turbo-preview* resulted in superior responses for the assessment task in stage four. This also highlights the importance of using evaluation metrics. In the testing phase, the metrics provide robust support for model choice, beyond our subjective review of the responses. In the application phase, the metrics flag to us which responses are reliable based on the information and instructions given, and which are not.

4 Results

From the ORIENTING S-LCA methodology, which encompassed 28 social topics, 10 were chosen for evaluating the automated pipeline's effectiveness compared to manual assessments. The selected topics were child labor, community engagement, contribution to economic development (including local employment), fair competition, health and safety (workers), prevention and mitigation of armed conflicts, remuneration and social benefits, respect of intellectual property rights, skill development and technology development, and women's empowerment. These topics were selected to cover a spectrum of complexity, with some being relatively straightforward to detect and others more subtle. Additionally, the selection process considered the ease of reaching a consensus on the level assessments among experts and practitioners, with some topics having achieved quick agreement and others necessitating more in-depth discussions for a unified conclusion in the manual assessment. The purpose of this selective approach was to gain a comprehensive insight into the AI-augmented tool's capability to handle varying degrees of difficulty and uncertainty in assessing social topics (Zanchi et al. 2023). Empirical data for the testing was gained from an S-LCA study, which was conducted for a beverage carton product as part of the

ORIENTING project. A more detailed description of the S-LCA case study is available in (Pihkola et al. 2024).

4.1 Manual S-LCA

The manual version of the S-LCA was conducted together with forest industry company Stora Enso, using a combination of primary and secondary data, collected for assessing the social performance of a typical beverage carton manufactured by Stora Enso. Researchers from VTT and a representative from Stora Enso conducted the manual case study together. The data and results included here represent selected parts (10 social topics and data collected for those topics) for some of the processes evaluated using the social performance assessment template (spreadsheet) developed in the ORIENTING project. The level score from this manual assessment for the selected 10 social topics tested in this study are shown in Table 2. The data used for the manual assessment of these 10 topics is described in Sect. 3.2.2. A table listing the URLs from which the data was collected, and a list of the social topic definitions and reference scale criteria for the 10 topics is given in the appendix.

Table 2 Manual assessment results by social topic

Social topic	Manual level
Child labor	1
Community engagement	2
Contribution to economic development (including local employment)	1
Fair competition	−1
Health and safety (workers)	1
Prevention and mitigation of armed conflicts	0
Remuneration and social benefits	2
Respect of intellectual property rights	0
Skill development and technology development	1
Women's empowerment	0

4.2 Automated S-LCA

Two hundred sixty-nine of 330 (81.5%) queries across 33 documents and 10 social topics returned responses that showed evaluation metrics marking them as valid responses within 3 attempts. Of these valid responses, there were 101 query responses that detected relevant information to the given social topic. For some documents, information relevant to multiple social topics was detected. Eight of the 10 social topics were detected among the full set of documents. Valid and relevant responses from each document and social topic query were then passed to the assessment stage.

Table 3 shows aggregated RAG Triad evaluation averages for the sets of document responses that were identified as valid and relevant for each social topic. These responses were then provided as input for the assessment query for each social topic. The table also provides the overall validity results for each social topic. "Valid" is the count of documents for which a valid response was given within three query attempts (a valid response can be "no information," however). "Relevant and valid" is the count of documents where the social topic query produced a valid response that was not "no information."

"Community engagement" had the lowest validity performance, with only 19 of 33 documents having a valid query response within three attempts. "Child labor" had the highest validity performance, with only one invalid document response. The differences in validity performance between the social topics are likely due to variations in the conceptual

Table 3 Aggregated RAG Triad evaluation statistics for relevant and valid document responses by social topic

Social topic	Count		Average			Evaluation average
	Valid	Relevant and valid	Answer relevance	Context relevance	Groundedness	
Child labor	32	8	0.81	0.78	0.7	0.76
Community engagement	19	11	0.67	0.75	0.61	0.68
Contribution to economic development (including local employment)	26	19	0.8	0.71	0.5	0.67
Fair competition	29	5	0.64	0.68	0.6	0.64
Health and safety (workers)	24	20	0.75	0.79	0.51	0.68
Prevention and mitigation of armed conflicts	30	0	NA	NA	NA	NA
Remuneration and social benefits	27	9	0.62	0.71	0.5	0.61
Respect of intellectual property rights	27	8	0.84	0.81	0.65	0.77
Skill development and technology development	31	21	0.8	0.57	0.52	0.63
Women's empowerment	24	0	NA	NA	NA	NA
Total	269	101	0.76	0.71	0.55	0.67

nuance of each topic, which manifests in the topics' definitions and reference scale indicators (for a list of the definitions and reference scales used in this analysis, see the appendix). Social topics with more straightforward, less nuanced definitions and which are more likely to be disclosed in a straightforward manner in the documents are therefore easier to detect and detect consistently (like child labor). Meanwhile, other topics that require a more careful interpretation of the definition against information that may be disclosed in a more passive manner in the documents (like community engagement) result in less consistent and straightforward responses, which is reflected in the evaluations.

"Contribution to economic development (including local employment)," "health and safety (workers)," and "skill development and technology development" had relevant and valid responses among a high number of documents, 19, 20, and 21, respectively. The average Answer Relevance, Context Relevance, and Groundedness is derived across the evaluation metrics of the relevant and valid responses. The "evaluation average" column averages the Answer Relevance, Context Relevance, and Groundedness columns. The "remuneration and social benefits" social topic shows the lowest average of evaluation metrics at 0.61, and "respect of intellectual property rights" shows the highest at 0.77. The remuneration and social benefits topic has a low average Answer Relevance score relative to the other topics, which brings down its average across the evaluation metrics, as well. In our review and discussions of the results, this was attributed to the social topic being a combination of what could also be considered as two distinct topics, remuneration as one and social benefits as another. The low Answer Relevance score reflects responses that were constructed on information that was relevant to remuneration but not social benefits or vice versa. This comes from some documents containing information about one of these issues, but not the other. Meanwhile, the high evaluation score for respect for intellectual property rights reflects the straightforward nature of discussions and disclosures of intellectual property rights. The high Context Relevance score that this topic receives suggests that information about this topic in the documents was concentrated and easily matched to the definition given. Interestingly, the information gathered in these responses was evaluated as relevant to the definition of the social topic from the viewpoint of safeguarding companies' intellectual property rights, not individuals'. This is discussed in more detail in Sect 4.3 and Sect 5.

The LLM-produced assessment level for each social topic is shown in Table 4 along with the Answer Relevance and Groundedness metrics of the assessment query. For all responses, the RAG evaluation metrics were greater than or equal to 0.8 for Answer Relevance and greater than 0.4 for Groundedness.We can conclude from Table 4 this that the model performed well across the social topics, returning focused responses that were evidenced by the information provided.

As mentioned, not all social topics were detected across the documents. For two social topics, "prevention and mitigation of armed conflicts" and "women's empowerment," none of the documents with valid query responses were found to contain relevant information for the social topic, according to the LLM. "Prevention and mitigation of armed conflicts" had three of the 33 document responses return as invalid across attempts. However, the LLM did not detect relevant information in any of the 30 valid responses. The source that was cited in the manual assessment for this social topic had a valid response, but no relevant information was detected by the LLM. "Women's empowerment" had nine of the 33 document responses return invalid across attempts. Of the three documents cited in the manual assessment of this social topic, one document was the one that could not be retrieved, another was evaluated as invalid, and the final was evaluated as valid, but no relevant information was detected. As a result, the LLM could not produce a level assessment for either of these two topics.

Table 4 Automated assessment results by social topic

Social topic	LLM level	Answer relevance	Groundedness
Child labor	1	0.80	0.93
Community engagement	2	1.00	0.88
Contribution to economic development (including local employment)	1	0.80	0.57
Fair competition	0	0.80	0.76
Health and safety (workers)	1	0.80	0.47
Prevention and mitigation of armed conflicts	NA	NA	NA
Remuneration and social benefits	0	0.80	0.42
Respect of intellectual property rights	2	1.00	1.00
Skill development and technology development	1	0.80	0.62
Women's empowerment	NA	NA	NA

4.3 Comparison

The LLM and manual level assessments are shown side by side for each social topic in Table 5. For 5 of 10 (50%) social topics, the LLM produced the same level as the manual assessment did. For 2 topics, the LLM was unable to produce a level due to a lack of input. For the final 3 topics, the LLM produced a different level than the manual assessment. For "fair competition," the LLM assigned level 0 while the manual assessment assigned level −1. For "remuneration and social benefits," the LLM assigned level 0 while the manual assessment assigned level 2, and for "respect of intellectual property rights," the LLM assigned level 2 while the manual assessment assigned level 0.

There does not appear to be a pattern in the document level or assessment level evaluation results that would suggest a higher or lower likelihood of the LLM assessment resulting in a different level than the manual assessment. For example, referring to Table 3, "fair competition" and "remuneration and social benefits" show relatively low average metrics at the document level and the assessment level when compared to other social topics. However, the evaluation results for "respect of intellectual property rights" are relatively high. Overall, the model performs well across social topics, providing evidenced, objective assessments based on the information it has been given. This suggests that the discrepancies were not due to underlying model performance but instead arose from issues in the inputs and instructions provided in the assessment pipeline. A thorough review of the LLM and human evaluations revealed that this is indeed the case.

The review of the LLM and human assessments of "fair competition" showed that the human evaluator inherently weighed the weakest, or most negative, discovered evidence most heavily. However, prioritizing or weighting certain evidence was not included as instructions to the LLM in the queries applied. In the context of S-LCA, a single negative piece of evidence among many neutral or even positive evidence pieces should influence the assessment level, to avoid compensating for potential negative results in one process or life cycle stage by positive results in other processes or life cycle stages. Future exercises should consider incorporating explicit prioritization or weighting instructions in the LLM system prompt or actual query to explore how this would affect the alignment of the LLM responses with the manual assessment.

Table 5 LLM and manual level assessments by social topic

Social topic	LLM Level	Manual level
Child labor	1	1
Community engagement	2	2
Contribution to economic development (including local employment)	1	1
Fair competition	0	−1
Health and safety (workers)	1	1
Prevention and mitigation of armed conflicts	NA	0
Remuneration and social benefits	0	2
Respect of intellectual property rights	2	0
Skill development and technology development	1	1
Women's empowerment	NA	0

For "remuneration and social benefits," it was determined that the AI provided a more objective assessment than the human evaluator, based on the information available. Here, the manual assessment was positively biased in two ways. First, by the human evaluator's background knowledge about the Finnish employment context, regulations, and public services, and second, by verbal communication with company representatives. Thus, the scoring was partly based on knowledge and data that were not documented as part of the reference documents. Further, the social topic definition actually combined two different topics (social benefits and remuneration), while it was possible to do the scoring based on information collected for only one of those topics. This underlines the importance of clear definitions of social topics to be assessed, but also the usability of AI in detecting potential deficits in collected background data (lack of evidence). In future exercises, it could be useful to include in the LLM's input materials the operating countries' regulatory context through, e.g., public documentation of what conditions and benefits employers are required to provide to employees by law. However, informal information the researcher has access to via, e.g., conversations with stakeholders, remains a challenge.

A similar issue of missing background knowledge was found in the case of "prevention and mitigation of armed conflicts," which the LLM was unable to provide an assessment on. In this case, the document cited in the human evaluation was not detected as relevant in the AI evaluation. Correct evaluation of this document required contextual knowledge of the Russia-Ukraine war with respect to the actions that the broader landscape of companies sourcing raw materials from Russia were taking after the escalation of the war in 2022. During the manual case study, the original social topic definition had to be adjusted to become relevant for the context of the case study. These changes were acknowledged by the human evaluators, but not cited in the reference documents or reflected in the original social topic definitions.

Under "respect of intellectual property rights," it became clear that the AI assessment was performed from the perspective of the company as the stakeholder, while the human assessment was performed from the perspective of the worker as the stakeholder. The substantial difference

in responses here confirms the importance of performing S-LCA from multiple relevant stakeholder perspectives. At the time of this study, this had not yet been clearly incorporated into the ORIENTING S-LCA performance assessment template, though it has since.

5 Discussion

The 50% alignment rate between the AI and manual assessments in this study underscores the significant potential of LLMs to advance the efficiency, scalability, and objectivity of S-LCAs. However, the discrepancies in the remaining assessments highlight areas where the contextual knowledge and nuanced judgment that human evaluators currently bring to the table are required. Figure 3 summarizes findings of the comparative analysis, which benchmarks the AI evaluations in our case with the human-produced evaluations.

In cases where the evaluations aligned, the social topic descriptions and assessment criteria were straightforward, with less room for interpretation. The instructions provided to the LLM in query format were clear and explicit in the sense that they covered the requirements necessary to develop a consistent evaluation in the desired context. In contrast, the cases of misalignment were subject to specific pitfalls in the instructions and information provided that resulted in objective, but insufficient assessments, or insufficient documentation of applied background information. One of the primary factors contributing to the misaligned results is the reliance on contextual knowledge and implicit criteria by human evaluators. For instance, in the assessment of "remuneration and social benefits," the human evaluators' familiarity with the Finnish and Swedish employment context and their direct interactions with company representatives influenced their assessment. This context-specific knowledge was not accessible to the AI, resulting in an objective yet less contextually informed evaluation. In a similar vein, in the assessment of "fair competition," the human evaluator relied on implicit criteria derived from the broader context and purpose of the S-LCA, weighting negative evidence more heavily than neutral or positive evidence. Again, this context was not made explicit to the AI, resulting in different evaluation outcomes. The difference in results for "respect of intellectual property rights" was interesting because the evaluation outcomes were clearly a result of taking different stakeholder perspectives. This emphasized the importance of stakeholder perspective in S-LCA, as the assessment can vary dramatically across perspectives.

Importantly, these differentiating factors are not a result of poor LLM performance. This can be determined from the evaluation metrics, which show a range of decent to excellent performance with respect to the LLM's construction of an evidenced and objective response, based on the inputs and instructions provided. Instead, the differences arose from how the assessment was formulated and what the AI was instructed to do (or not instructed to do) in this pipeline. This is promising, as these issues are relatively easy to address in future use cases. For example, measures to better align the AI evaluation outcomes with the purpose and needs of the S-LCA could include incorporating detailed contextual materials and explicitly defining the criteria and desired nuances in the AI task instructions (e.g., defining evidence weighting and stakeholder perspective). This furthers a growing body of work demonstrating that LLMs can successfully support qualitative data workflows. Prior research has shown that LLMs can match or exceed human performance in consistency and efficiency when tasks are well-defined and appropriately scoped (Fischer and Biemann 2024; Polak and Morgan 2024; Bronzini et al. 2024). In this respect, our findings are consistent: when social topic assessments were framed with clear criteria and concrete input materials, the LLM performed reliably.

To our knowledge, this study presents the first practical demonstration of LLM-supported S-LCA, covering both data collection and reference-scale assessment, and therefore

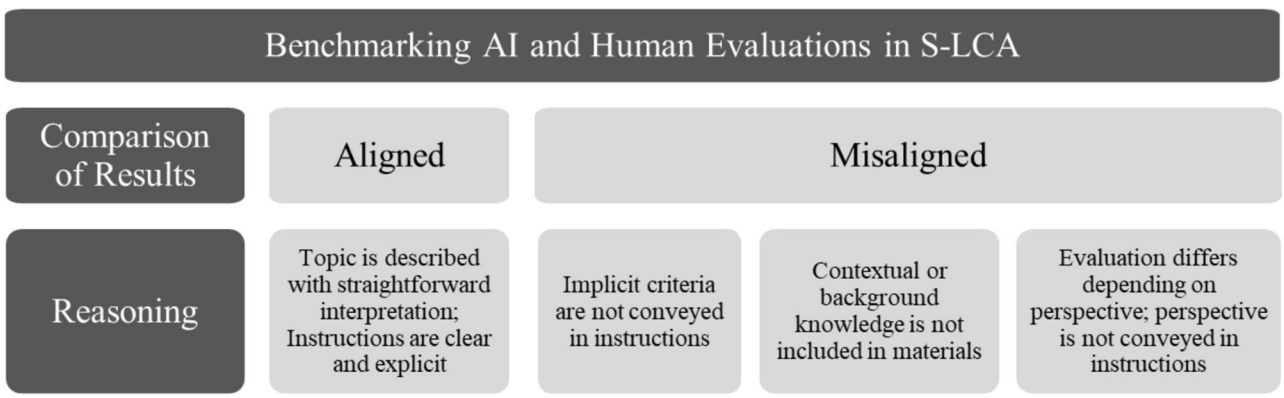


Fig. 3 Summary of findings from benchmarking AI with human evaluations

uncovers domain-specific challenges such as stakeholder sensitivity and interpretation of context-dependent social indicators. While prior work has emphasized LLM capabilities in contextual comprehension (Törnberg 2024; Anagnostopoulos et al. 2025), our results suggest that successful realization of these capabilities for social assessments requires careful and iterative prompt design, as well as inclusion of supplementary background data to provide necessary context for reference scale scoring. Recent research has also highlighted the risk of unintended bias in LLM-generated content, particularly in tasks that require specific perspectives or value judgements (Rask and Shimizu 2024). We reflect this concern, as our results show that stakeholder perspective and local contextual knowledge played a role in shaping human evaluations but were not specified in the LLM instructions, leading to divergence in the assessment of some social topics. Emerging research on the use of AI personas, constructed by fine-tuning or prompting an LLM to generate content based on a set of criteria, norms, and values representing a persona type or stakeholder group, shows promise here (Arora et al. 2024; Sun et al. 2024). Future research on the use of LLMs in S-LCA and other social research involving assessments that may be sensitive to stakeholder perspectives should consider this evolving field of research, as the use of AI persona techniques may help mitigate some of the limitations observed here. Additionally, potential techniques for using AI to support the cause-and-effect chain definition when using the impact pathway approach would be another interesting topic of future research in AI-augmented S-LCA.

Our comparative analysis between the AI-augmented and manual S-LCAs also contributes to a broader discussion about the effectiveness and limitations of LLMs for evaluation purposes. While LLMs can significantly enhance the efficiency and scalability of evaluations, without explicit instruction, they lack the nuanced understanding and contextual awareness that human evaluators bring. In many cases, this is not an insurmountable barrier, as this is information that can be conveyed either through the provision of additional context materials or carefully defining and refining of task instructions. However, the missing information required to perform a complex evaluation task in the manner desired can only be revealed through benchmarking exercises such as the one performed in this analysis. For instance, while the RAG evaluation metrics can inform us that the LLM has performed well in defining the assessment score for each of the 8 out of 10 social topics evaluated, it was only through comparison with a human evaluation that we were able to uncover areas of improvement for the automated process. At the same time, by comparing the human evaluation outcomes with the AI evaluation outcomes, we were able to detect assessment scores that were, in fact, reliant on subjectivity or weak evidence due to insufficient documentation of applied background knowledge and data. The alignment of half of the assessment scores demonstrates that AI can achieve considerable accuracy with clear and explicit instructions. Meanwhile, the misalignments show that comparison with human evaluations remains critical to further development of these methods.

These findings align with recent scholarship on the automated evaluation of LLMs and human-in-the-loop (HITL) approaches, which suggests that automated evaluations are well-suited to information retrieval tasks but advocates the importance of human oversight in more sensitive and critical judgment tasks (Shankar et al. 2024; Abeysinghe and Circi 2024). However, reliance on human judgement and human annotations as a gold standard or ground truth has also been criticized (Wang et al. 2020). In the future, a significant amount of time could be saved if an AI-augmented framework could be used for pre-scanning and detecting relevant information from available background documents and for detecting data gaps for different social topics. The time required for screening and reading various background documents from different sources is currently one of the bottlenecks hindering S-LCA application. By implementing a hybrid approach that utilizes LLMs and automated evaluations for information retrieval, while maintaining expert oversight for the more sensitive and critical tasks, the time of social experts could be saved and more wisely used in checking potential cases of conflicts and/or in confirming the provided scores from documents and text-extracts already identified as relevant. Further, while this study focused on tasks of information extraction and issue assessment, future research could expand this workflow to explore automation of data source curation, as well. The set of webpages used in this study is the same set of sources used in the manual S-LCA that we sought to benchmark and compare the automated method with. However, searching for and verifying relevant data sources is a time-consuming task for experts, and significant efficiencies could be gained by applying automation techniques here, as well. Such an endeavor would require robust quality checks of the webpages and data sources collected, using automated or manual evaluations or a hybrid approach.

6 Conclusion

This study presents a practical framework for integrating LLMs into a qualitative reference-scale S-LCA. It highlights the potential benefits and limitations of AI-augmented assessments and takes significant steps towards greater efficiency and scalability of S-LCA and similar assessment tasks. The AI method developed in this study achieves a 50% alignment rate of social topic scores when compared with the parallel human evaluation. A thorough analysis of the misaligned scores reveals areas of improvement for the AI method and cases in which human evaluation fell short of the AI.

By benchmarking the AI evaluation outcomes with human evaluation outcomes, we uncover the main factors contributing to the evaluation misalignments and present avenues for improvement. Beyond this case study, we suggest, for constructions of AI methods aimed at complex social evaluation, an approach that utilizes human benchmarking in the development process to produce mutual learnings. It is not enough to rely on metrics evaluating the objective performance of the model in the assessment pipeline. These metrics are necessary to ensure the accuracy and reliability of the LLM's output based on the input and instructions provided and should be interpreted as such. Benchmarking with a human-produced assessment is needed to properly evaluate whether the input and instructions provided are sufficient to reach a valid assessment in the evaluation context. Also, to consider the constantly changing operating environment and the geographic, economic, regulatory, and cultural context in which the assessment is conducted.

The insights gained from this research are valuable for stakeholders, researchers, and policymakers in the ongoing development and adoption of AI-augmented approaches in S-LCA and other evaluation contexts that require a blend of objective assessment and contextualized judgment. Future research would benefit from testing AI-assistance pipelines such as the one developed in this study in diverse industry contexts. A limitation of this study is the application of the AI method to only one case study, out of which a limited number of social topics were selected for this test case. While this application has progressed our understanding of the benefits and limitations of employing AI in assessment tasks, further development of the method based on these findings is needed. Applications with different S-LCA methods and impact assessment approaches, conducted in different industry contexts using more versatile data sources, would likely reveal additional areas for refinement of the method for S-LCA.

Appendix

Social topic definitions and reference scale definitions and indicators used in this study (extracted from Zanchi et al. (2023))

Child labor

WHAT THE SOCIAL TOPIC IS ABOUT: Child labor is work that deprives children of their childhood, their potential, and/or their dignity, and that is harmful to their physical and/or mental development. In its most extreme forms, child labor involves children being enslaved, separated from their families, exposed to serious hazards and illnesses, and/or left to fend for themselves on the streets.

WHAT WE WANT TO MEASURE: The extent to which a company works towards eradicating child labor and pro-actively raises awareness of issues associated with child labor. The assessment aims to verify if the organization might or is employing children (as defined in the ILO conventions), if the conditions are favorable for the occurrence and/or eradiation of child labor, and the existence and quality of the prevention and mitigating measures taken by the organization.

REFERENCE SCALE: Level – 2: Incidents of child labor, including the worst forms of child labor and children in Hazardous work, have been discovered. Indicators: Absence of positive information while the company is in an area where this situation often occurs according to statistics NGOs reports, voices from local communities, or local schools that indicate child labor.

Level – 1: Incidents of child labor but no incidents referring to the worst forms of child labor or children in Hazardous work have been discovered within the company or facility, and a corrective action plan with a clear timeline for completion has been developed. Indicators: While the company is in an area where this situation often occurs according to statistics, there is evidence that the company has started to address the situation with a clearly defined timeline presence of organization documents, Interview with directors or human resources officer that indicate no worst forms of child labor, and/or a corrective action plan.

Level 0: No incidents of child labor are discovered, and the company or facility has a management system in place that enforces the policy prohibiting child labor. Indicators: Records proof of age upon recruitment, including copies of documents such as birth certificates, passports, or religious, medical, or other records. The company is not in an area where child labor is likely, and there is no reason to suspect child labor.

Level 1: The company or facility has a management system in place to raise awareness of issues associated with child labor within its organization, its subcontractors and its tier 1 suppliers, with the aim to address the root courses of child labor in the region it operates, and takes action as appropriate, like improving the accessibility to schools. Indicators: Evidence that describes the management system of the organization, its subcontractors, and its tier 1 suppliers is effective, for instance, by embedding several of the following examples: Amended job descriptions to include responsibility for children's rights. Staff are trained, empowered, and incentivized to deliver on child rights goals. There is a sensitive grievance mechanism for child rights issues. There is an ombudsperson for children.

Level 2: The company has made eliminating the use of child labor in its own organization, its entire value chain, including clients, a top priority, and can demonstrate the success of its approach. Indicators: Presence of a child labor project related with detailed informations, monitoring data about pre and post situations in its own organization and along supply chain.

Community engagement

WHAT THE SOCIAL TOPIC IS ABOUT: Community engagement concerns any form of relationship between the company and community stakeholders (individuals or community groups) that may be affected by the implementation of business policies and actions of an organization that affect the local environment, health, and well-being. Different ways of engagement could exist such as the inclusion of community members representatives in decision-making processes, or direct involvement in community initiatives, or financial support of community projects (e.g., Earth Day activities, recycling initiatives, and visits to local schools). Particular attention needs to be paid to engaging representatives of vulnerable groups such as indigenous peoples, youth, and women.

WHAT WE WANT TO MEASURE: Whether, and to which extend, the company or facility includes community stakeholders in relevant decision-making processes through ongoing open dialogue and responds to their concerns and inquiries fairly and promptly, to continuously foster greater trust and relationship with the local community. The Aarhous Declaration distinguishes 5 levels: Inform (We will inform), Consult (we will inform and listen, but not more), Involve (We will provide feedback how your input influenced our decision), Collaborate (We will seek your advice and use it to the maximum extend possible), Empower (We will implement what you decide).

REFERENCE SCALE: Level – 2: The company and facility do not make any efforts to inform or consult the communities, and have a strained relationship with the local communities. Indicators: Absence of positive information while the company is in an area where this situation often occurs, according to statistics.

Level – 1: The company and facility do not make serious efforts to inform or consult the communities, but the relationship with the local communities is normal. Indicators: Some complaints but no fundamental criticism from NGOs and others about the behavior of the company.

Level 0: The company or facility has a system or mechanism in place to enforce the policy to inform and consult the local community. Indicators: Documents and statements that reveal the efforts and results that show the policy at least establishes formal communication channels between the company or facility and the community to inform then and to hear the opinions and views (e.g., minutes or registration of meetings, presence of information/communication materials online-offline, presence of a dedicated office/person).

Level 1: The company or facility has a system or mechanism in place to enforce the policy to inform, consult, and involve the local community. Indicators: The company or facility has a system or mechanism in place to enforce the policy to inform, consult, and involve the local community. Diverse group of communities are included in the engagement that takes place frequently enough (e.g., minutes or registration of the meetings). Information obtained from NGOs and other credible sources shows actions and results. Company self-declaration containing clear examples and an overview of objectives, actions, and results obtained.

Level 2: The company or facility has a system or mechanism in place to enforce the policy to inform, consult involve and collaborate with the local community, (see definitions) Indicators: The company has implemented rules that require them to base decisions on local community consent if such decisions have serious implications for the community. There is evidence that the company or facility has taken the inputs from community dialogues seriously and, where appropriate, has softened the consequences of the decision. The company or facility pays special attention to the voice of vulnerable groups, such as indigenous peoples and women. The company provides support (volunteer-hr or financial) for community initiatives. Presence of a clear and structured plan and a dedicated working group including both representatives of the company and of diverse community stakeholder groups.

Contribution to economic development (including local employment)

WHAT THE SOCIAL TOPIC IS ABOUT: There are several definitions of economic development; in this context, it is defined as "the cultivation of activities that create a net gain of money into the community." Organizations can foster economic development in many ways, directly or through supply chain actors. They can generate revenue, create jobs, provide education and training, make investments, or forward research. Many social topics defined in this list are addressing parts of the puzzle needed to bring economic development; this social topic is particularly focused on the creation of jobs and generating income for local communities.

In addition to that, poverty alleviation and wealth distribution targets can be included in this topic. Poverty alleviation is in line with the first Sustainable Development Goal and aims to "end poverty in all its forms everywhere." Priority actions on poverty eradication include: improving access to sustainable livelihoods, entrepreneurial opportunities and productive resources; providing universal access to basic social services; progressively developing social protection systems to support those who cannot support themselves; empowering people living in poverty and their organizations; addressing the disproportionate impact of poverty on women; working with interested donors and recipients to allocate increased shares of ODA to poverty eradication; and intensifying international cooperation for poverty eradication (https://sdgs.un.org/goals/goal1). Wealth distribution is in line with the Sustainable Development Goal 10, aiming to "reduce inequalities."

WHAT WE WANT TO MEASURE: Whether and the extent to which the company or facility contributes to the economic development of the local community, to the poverty alleviation, and toward wealth distribution in the local community and society in general.

REFERENCE SCALE: Level − 2: The company deliberately does not support, or damages the attempts of local authorities to create the conditions for economic growth, engaging in tax evasion, corruption, and other detrimental policies or actions Indicators: Absence of positive information while the company is in an area where this situation often occurs according to statistics.

Level − 1: The company, until recently, deliberately did not support or damaged attempts from local governments to facilitate economic growth, but has changed its attitude and has implemented a policy to end corruption, tax evasion, and other forms of malpractice with a clearly defined timeline. Indicators: There are incidents of complaints, lawsuits, and other signals, but they have been significantly reduced during the last 3 years.

Level 0: The company employs local workers and sources from local communities purely based on business criteria like quality, price, and timing, and does not engage in actions that can weaken the local governance mechanism. Indicators: Absence of complaints, Information obtained from unions, NGOs, and locals confirming this.

Level 1: The company has a policy to stimulate job creation in the local community by hiring new staff, working with local suppliers or sub-contractors. The policy includes a commitment to help increase the economic growth in the region. Indicators: Description of the policy by the company and signs that this policy is working, Comments from NGOs and locals, including local governments.

Level 2: Indicators: Reports or publications that describe the public–private partnership and the investments. Examples of public–private partnerships provided by the company, Documentation showing how the company helps smallholders operating in a loose value chain to become a commercial supplier on markets from which others can source.

Fair competition

WHAT THE SOCIAL TOPIC IS ABOUT: Fair competition concerns competitive activities that are conducted in a fair way and in compliance with legislations preventing anti-competitive behavior, anti-trust, or monopoly practices. Fair trading mainly applies in the condition of commercial relationship between companies differing in terms of size, and economic and commercial development capacity and opportunities (e.g., small-scale entrepreneurs).

WHAT WE WANT TO MEASURE: Whether the organization's competitive activities are conducted in a fair way and the extent to which a company considers the potential impacts or unintended consequences of its procurement and purchasing decisions on other organizations, and take due care to avoid or minimize any negative impact.

REFERENCE SCALE: Level − 2: Evidence indicates that there is a substantial risk of unfair competition. The company sourcing in this area is not take actions or is even complicit in this process. Indicators: Complaints, lawsuits, and other signs that may indicate issues. Absence of positive information, while the company is in an area where this situation often occurs, according to statistics.

Level − 1: The company shows several incidents of complaints, lawsuits, and other signals, but they have been significantly reduced during the last 3 years. Indicators: Interviews with community members, employees, governmental agencies, union branches, management, and NGOs. There are still complaints, lawsuits, and other signs, but they have been reduced considerably in the last 3 years.

Level 0: The company presents credible documents demonstrating that the company insists on maintaining fair business relationships in its internal procedures or that it works under a relevant certification scheme. Indicators: Credible documents and statements from third parties or the company that show the contribution of the company in awareness raising and training regarding the establishment of cooperatives or making local buyers aware of the requirements set by the company. Credible documents that show that the company insists on the implementation of fair competition in its internal procedures. The community works under a relevant certification scheme.

Level 1: The company has credible documents and statements from third parties that confirm that the company's policy is effective in translating this commitment into real action in the interest of fair competition. Indicators: Credible documents and third party statements confirming that the company's policy is effective in translating this commitment into real actions in translating this commitment into real actions. Credible statistics that the business relationship in this community is indeed improving and better than average, and that this can be at least partially community is indeed improving and better than average and that this can be at least partially linked to the company's efforts.

Level 2: The company or facility has publicly stated that a fair trade relationship with the communities it sources from is a high priority and publicly reports and discloses its management system/initiatives/activities commitments, performance, progress, and effectiveness. Indicators: The company has invested in and engaged with a local public–private partnership to improve the trading relationship in the region, and the partnership can demonstrate significant progress in the last 3 years. Statements by NGOs, local authorities, credible newspapers, awards, and others, with similar content.

Health and safety (workers)

WHAT THE SOCIAL TOPIC IS ABOUT: This topic includes all the management practices and strategies the organization should guarantee in terms of health and safety of workers, consumer, local community, and small-scale entrepreneurs.

The main focus in occupational health&safety for workers and small-scale entrepreneurs is on three different objectives: (i) the maintenance and promotion of workers' health and working capacity; (ii) the improvement of working environment and work to become conducive to safety and health and (iii) development of work organizations and working cultures in a direction which supports health and safety at work and, in doing so, also promotes a positive social climate and smooth operation and may enhance productivity of the undertakings. We interpret this also to include protection against all forms of harassment, including sexual harassment.

Consumer health and safety refers to the consumers' rights to be protected against products and services that may be hazardous to health or life (ISO 26000, 2008). Customers (end users) expect products and services to perform their intended functions satisfactorily and not pose a risk to their health and safety.

Health and safety for local communities includes any actions, related to the work activities, that could produce positive or negative impacts for the surrounding (e.g., accidents related to unsafe equipment or structural failures of buildings; project-related land use changes can also lead to natural disasters such as landslides). In the case of the local community, also the concept of secure living conditions needs to be considered. Organizations with weak security oversight may contribute to insecure living conditions, community tensions, and regional conflicts. At the same time, organizations that enter relatively insecure regions may improve living conditions through the responsible actions of private security personnel. Organizations may employ security forces to protect their employees and assets. This security should extend to the protection of human rights in surrounding communities.

WHAT WE WANT TO MEASURE: The extent to which the management maintains or improves the safety and overall health status of the workers, local communities, and small-scale entrepreneurs; the extent to which the product, under defined conditions maintains or improves the health status and safety of the users in the target market.

REFERENCE SCALE: Level − 2: There is neglect in the health and safety conditions and working culture of stakeholders, and incidents of high accident rates lead to deteriorating conditions (in accordance with the global burden of disease as monitored by the World Health Organisation). Significant damage occurs with either no action taken by the company or complicit inaction. The normal use of the product has negative health or safety impacts on the user on the short or long term. The rate of incidents and status of prevention measures and management practices are the focus. An incident is defined as a work-related event in which an injury or ill health or fatality occurred or could have occurred. Source: Black text adapted from PSIA (2022), red written by PRE based on UNEP Methodological sheet (2021) and description in cell J16 Indicators: A high number/percentage of incidents, complaints, lawsuits, or hours of injuries per level of employee. Absence of positive information to address health and safety issues where the risk on stakeholders is high in accordance with generic statistics.

Level − 1: There is neglect in the health and safety conditions and culture of stakeholders, and incidents of high accident rates lead to deteriorating conditions (in accordance with the global burden of disease as monitored by the World Health Organization). Damage occurs, but the company has developed a corrective plan and policy with a clear timeline for completion. The normal use of the product has negative health or safety impacts, but the producer has a corrective plan to reduce the problem. Indicators: Presence of a formal policy concerning health and safety, and adequate measures taken, with evidence from reports showing improvement. While the company is in an area where this situation often occurs, according to statistics, there is evidence that the company has started to address the situation with a clearly defined timeline. Complaints and reports indicating malpractice have reduced during the last 3 years.

Level 0: Working conditions and working culture are adequately protecting occupational health and safety, which includes that equipment, the use of personal protection equipment, the prevention of harassment are conforming to the state of the art regarding safety and exposure. The normal use of the product does not have any detrimental effect on the user. Indicators: Documents like audits that show compliance with national standards, see for example the Global ILO LEGOSH database for OECD countries. Documents that show certification schemes/standards on health and safety, audits. Presence of formal policies concerning health and safety, and preventive measures.

Level 1: The company has a management system in place to pro-actively and continuously improve the working culture, beyond an acceptable level, and can show tangible results of these efforts. There is evidence to suggest that the product is successfully designed to create a maximum contribution to the health and safety of the intended user Indicators: Documents that provide a credible description of management system to promote continuous improvement of health and safety and the results of these efforts Credible statements from NGOs, unions and workers that confirm this A very low number or percentages of injuries and fatal accidents compared to statistics of similar industries/companies.

Level 2: The company is best in class compared to its peers on OHS performance There is solid science-based evidence that the normal use of the product significantly contributes to health and safety of the user and reaches vulnerable groups who could benefit from it Indicators: Credible statistics show the OHS performance is best in class compared to its peers in the same sector and region, and this performance has improved over at least 3 years Credible statements from NGOs, unions and workers that confirm this A very low number or percentages of injuries and fatal accidents compared to statistics of similar industries/companies.

Prevention and mitigation of armed conflicts

WHAT THE SOCIAL TOPIC IS ABOUT: Prevention and mitigation of armed conflicts is one of the target of the Sustainable Development Goal 16 "peace, justice and strong institutions." Conflicts can be defined as a tense situation between different parties caused by different interests, aims, or value systems. Armed conflict is a dynamic process of violent confrontation between two or more antagonists. There are special regions in the world that are known for enduring disturbances—so-called conflict zones. This subcategory shall also consider if the organization acts in conflict zones.

WHAT WE WANT TO MEASURE: This topic assesses the organization's role in conflicts or situations that might in the future develop into conflicts. Thereby, both positive and negative impacts on conflict developments are taken into account.

REFERENCE SCALE: Level − 2: The company plays a role in the development of conflicts through its operations, practices, or influence in an area of violent conflict. The company, though aware, takes no action to address the issue, withdraw its practices, or engage with the stakeholders affected. Indicators: Interviews with community members and NGOs reports by authorities indicating incidents of spurred conflict. An absence of positive information indicating the company is addressing the conflict situation.

Level − 1: The company plays a role in the development of conflicts through its operations, practices, or influence in an area of non-violent or violent conflict. The company has taken steps in addressing developing a strategy to reduce and prevent conflicts in the area of the conflict zone in which it operates or its supply chain operates. Indicators: Interviews with community members and NGOs reports by authorities indicating incidents of spurred conflict, but substantiation of the issue being addressed through actions by the business that are recognized by the appropriate authority.

Level 0: The company adheres to the standard on international humanitarian law, thus helping to avoid the risk of causing or contributing to adverse impacts when operating directly or indirectly (through its supply chain) in conflict zones (as defined by the Heidelberg Institute for International Conflict Research). Indicators: Internal report indicating that the company does not operate in non-violent or violent conflict zones.

Level 1: The company has put in place a strategy, measures, and/or an action plan to reduce and prevent conflicts when in operates in conflict zones or its supply chain operation in conflict zones. Indicators: Reports from authorities indicate that the company has put in place a strategy, measures, and/or an action plan in conflict zones.

Level 2: The company has put in place a strategy, measures, and/or an action plan to reduce and prevent conflicts when in operates in conflict zones or its supply chain operation in conflict zones. The company takes measures to raise awareness through information campaigns to draw attention from the parties involved to engage in dialogue, exchanges, and analyses towards confronting the consequences of conflict. Indicators: Reports from authorities indicate that the company has put in place a strategy, measures, and/or an action plan in conflict zones.

Remuneration and social benefits

WHAT THE SOCIAL TOPIC IS ABOUT: Remuneration means compensation for workers in terms of salary and social benefits. Salary should be fair, that is, a wage fairly and reasonably commensurate with the value of a particular service or class of service rendered, and, in establishing a minimum fair wage for such service or class of service. Codes of conduct which deal with wages and benefits rely on the following concepts: National Poverty Line (minimum income level on which an individual is supposed to be able to survive); Minimum wage (wage established by law, collective bargaining agreement or industry standard); Living wage (can be calculated according to different methods). Social benefits refer to non-monetary employment compensation as far as offered by the employer (while in many countries, such services are also offered by the government). Four basic categories of social Security benefits can be included in workers' earnings: retirement, disability coverage, healthcare and, in case of termination of employment, severance pay. Other social benefits that may be provided include medical insurance, dental insurance, paramedical insurance, including preventive medicine, medicine insurance, wage insurance, paid sick leave, paid maternity and paternity leave (parental leave is covered in work-life balance). Also, employment relationship, intended as the legal link between employers and employees that should enable

the application of laws related to social security and other labor laws, could be considered within this topic.

WHAT WE WANT TO MEASURE: The extent to which the management sufficiently compensates the workers in terms of salary and social benefits, and in compliance with established standards and if the wage provided is meeting legal requirements, whether it is above, meeting or below industry average and whether it can be considered as a living wage.

REFERENCE SCALE: Level − 2: Workers are paid below the poverty line in the country or region, or if this is not defined in the country or region, the payment is clearly insufficient, or workers are not paid the legal or industry minimum wage, and/or social benefits are not according to applicable law. There is no commitment to address this issue. Indicators: Complaints, lawsuits, and other signals that may signal issues. Absence of positive information while the company is in an area where this situation often occurs, according to statistics.

Level − 1: Not all workers are paid the living wage for a single household, or are not paid the legal or industry minimum wage, and/or social benefits are not according to applicable law, but the company has committed to resolve this issue clearly defined timeline. Indicators: While the company is in an area where this situation often occurs, according to statistics, there is evidence that the company has started to address the situation with a clearly defined timeline. The number of complaints, lawsuits, and other signals has been significantly reduced during the last 3 years.

Level 0: All workers are paid the living wage for a single household. Indicators: Audited statements and other credible statements from the company, or information obtained from unions, NGOs, and local workers providing evidence that this condition is fulfilled.

Level 1: All workers are paid at least a living wage for a standard family as defined by the methodology of the Wage indicator. Indicators: Audited statements and other credible statements from the company or information obtained from NGOs, unions, and workers, showing the gross wage is equal or higher than the living wage, as published by the government or as found in sources such as the Wage-indicator. In countries where the minimum wage is higher than the living wage, it can be sufficient to certify that the minimum wage is paid.

Level 2: All workers receive the (major share of) additional social benefits as defined for retirement, health insurance, disability coverage on top of the living wage levels for a standard family defined in the Wage-indicator project. Indicators: Audited statements and other credible statements from the company or information obtained from NGOs, unions, and workers, showing the gross wage is equal or higher than the living wage, as published by the government or as found in sources such as the Wage-indicator, plus a reasonable coverage of the social benefits.

Respect of intellectual property rights

WHAT THE SOCIAL TOPIC IS ABOUT: Intellectual property rights refer to the general term for the assignment of property rights through patents, copyrights, and trademarks. These property rights allow the holder to exercise a monopoly on the use of the item for a specified period.

WHAT WE WANT TO MEASURE: This subcategory assesses whether the organization's actions safeguard and value the creators and other producers of intellectual goods and services. The legal rights dealing with the intellectual property entail intellectual activities in the industrial, scientific, literary, and artistic fields.

REFERENCE SCALE: Level − 2: The actions of the company have resulted in violations of intellectual property rights, and a corrective action plan with a timeline for completion has not been developed. Indicators: Complaints, lawsuits, and other signals that indicate issues. Absence of positive information while the company is in an area/works in a field where this situation often occurs, according to statistics.

Level − 1: The actions of the company have resulted in violations of intellectual property rights, but a corrective action plan with a timeline for completion has been developed. Indicators: The situation has been unacceptable, but information obtained from NGOs and other credible sources showing the policy is working. There are incidents of complaints, lawsuits, and other signals, but they have been significantly reduced during the last 3 years.

Level 0: No violations of intellectual property rights that can be related to actions by the company have been discovered in the last 3 years. Indicators: No complaints, lawsuits, and other signals from the past 3 years.

Level 1: The company or facility has an effective management system in place to ensure that intellectual property rights are respected. Indicators: Information obtained from NGOs and other credible sources showing the policy is working. Company self-declaration containing clear examples and an overview of the results.

Level 2: The company or facility has an effective management system in place to ensure that intellectual property rights are respected, which has resulted in real improvements. Indicators: Documents and statements that reveal the efforts and results, such as reviews and opinions of local NGOs, verifiers, and other credible sources. Legal procedures initiated or actively supported by the company show its engagement.

Skill development and technology development

WHAT THE SOCIAL TOPIC IS ABOUT: Skill development concerns 4 categories: 1. Skills gap, the distance

between the skill level in society and the skills required to do a job in the company; 2. Skill obsolescence, the loss of skills due to the lack of use, or the risk the skills become irrelevant; 3. Skill shortages, when there are jobs, but no qualified staff in the community; 4. Over and under skilling, when people have skills above or below the job requirements. Skill development affects the community at large to creates a more resilient and healthy community, and potentially creates a resource for companies that look for new staff when needed. Technology transfer and development is the process for converting research into economic development, or for using technology, expertise, or know-how for a purpose not originally intended by the developing organization. It is fundamental between more advanced economies and developing economies for the improvement of social conditions and to prevent further environmental damage related to old technology use.

WHAT WE WANT TO MEASURE: The extent to which the company contributes to skill development for the community at large, not only in connection to the potential future staff needs but in general to produce benefits for all members of the community. Moreover, this social topic assesses whether the organization participates in joint research and development for efficient and environmental sound technologies.

REFERENCE SCALE: Level − 2: There is a significant skill-gap between the future needs of the company and the skill levels of local community members. The company is not doing anything to improve this situation. The company is not engaged in any environmentally sound technology development and transfer projects and does not plan to invest in such projects. Indicators: Complaints and other signals. Complaints from indigenous communities that they are discriminated against, when they apply for training. Absence of positive information while the company is in an area where this situation often occurs, according to statistics. For technology development: Interviews, company, and project reports.

Level − 1: There is a significant skill-gap between the future needs of the company and the skill levels of local community members, but the company has started to address this with an action plan with a clear timeline. The company has some on-going environmentally sound technology development and transfers programs and investments planned for the future, but they do not apply yet. Indicators: The skill level gap is significant, but information obtained from NGOs and local community voices shows the policy is working. For technology development: Interviews, company, and project reports.

Level 0: The company is managing the skill gap in a way that members of the local community are sufficiently qualified when new staff are hired. The company has environmentally sound technology development and transfer programs and collaborates with partners. Investments to technology development are made. Indicators: Reports from NGOs and others. Procedures disclosed by the company for technology development: Interviews, company and project reports, and funding providers' annual reports.

Level 1: The company actively contributes to reducing the skills mismatch by offering skill development for a relevant share of members of the local community. The company carries out environmentally sound technology development and transfer programs and invests heavily to them. New partnerships in research and development are actively sought. Indicators: Reports from NGOs and others. Examples of training materials provided by the company or others. Procedures disclosed by the company. For technology development interviews, company and project reports, and funding providers' annual reports.

Level 2: The company actively invests in reducing the skills mismatch in the region, and invests in a public–private partnership or invests in other activities that significantly increases training capacity and quality in the region for most members of the local community, not specifically guided by the companies own needs. The company carries out high ambition environmentally sound technology development and transfer programs, and invests heavily to them. There is a continuous search for more of such programs and research partnerships, and the company considers such activities a key point of its strategy. National and community development is a key priority. Indicators: Reports or publications that describe the public–private partnership and the investments. Documentation stating who can access the education and that the conditions are affordable for the intended local community. Statements from community members that confirm that most community members could attend and that they are applying or are ready to apply the skills they learned. For technology development: interviews, company and project reports, and funding providers' annual reports.

Women's empowerment

WHAT THE SOCIAL TOPIC IS ABOUT: The position of women is rated according to 4 dimensions: discrimination in the family; restricted physical integrity; restricted access to productive and financial resources; restricted civil liberties. We recognize that companies sourcing from small-scale entrepreneurs could influence only the third dimension "restricted access to productive and financial resources."

WHAT WE WANT TO MEASURE: The extent to which a company sourcing from a community of small-scale entrepreneurs is contributing to the empowerment of female small-scale entrepreneurs and the woman (spouses, daughters, etc.) related to male small-scale entrepreneurs.

REFERENCE SCALE: Level − 2: Evidence indicates that there is a substantial risk that women are discriminated and exploited. The company sourcing from this area does not take action or is even complicit in this process. Indicators: Complaints, lawsuits, and other signals that may signal issues. Absence of positive information, while the company is in an area where this situation often occurs, according to statistics.

Level − 1: Evidence indicates that there is a substantial risk that women are discriminated and exploited, but the company sourcing from this community, is actively supporting women groups and NGOs to stop this practice and empower them. Indicators: While the company is in an area where, according to the OECD SIGI database and the OECD regional studies, the SIGI score is above 0.25. Procedures have been disclosed by the company, together with some examples of cases where the company provided support. The situation has been unacceptable, but information obtained from NGOs and small-scale entrepreneur representatives shows the policy is working.

Level 0: The role of women is recognized, and the women rights are at an acceptable level. Indicators: According to the OECD SIGI database and the OECD regional studies, the SIGI score is below 0.25 in the country or region, and there are no signs the company is not behaving as the average. Reviews and opinions of local NGOs, verifiers, and other credible sources. Relevant certification from a credible NGO is available.

Level 1: The company that sources from this supplier's community has developed a policy and activities to further improve the access to services for female entrepreneurs, and it can report significant progress in the empowerment of women. Indicators: Reports or opinions from NGOs and others that describe the actions and investments that aim to improve the position of women. The company organizes awareness sessions and has implemented tools like a speak-up line etc.

Level 2: The company that sources from these suppliers has made the empowerment of women in the supply chain a high priority. This has resulted in a situation where women have equal, or at least sufficient, access to land, non-land resources, financial services, and have equal workplace rights. Indicators: Reports or opinions from NGOs and others that describe the actions and investments that aim to empower women. The company has invested in and engages in a public–private partnership to empower women in the supplier's community.

Table 6 High performing LLMs active in March 2024

Model	Provider	Release date	Context window (Tokens)
GPT-3.5 Turbo 16k	OpenAI	June 2023	16,384
GPT-4	OpenAI	March 2023	8192
GPT-4 Turbo	OpenAI	November 2023	128,000
Claude 2	Anthropic	July 2023	100,000
Claude 3	Anthropic	March 2024	200,000
Gemini Pro	Google	December 2023	32,768
Gemini Ultra	Google	February 2024	32,768
Llama 2 (70B)	Meta	July 2023	4096
Mistral 7B	Mistral AI	September 2023	8192
Mixtral 8×7B	Mistral AI	December 2023	32,768

Table 7 Links to analysis material

https://www.storaenso.com/fi-fi/newsroom/news/2019/7/investing-in-communities
https://www.storaenso.com/-/media/documents/download-center/documents/sustainability/storaenso_human_rights_guidelines_2021.pdf
https://www.storaenso.com/en/newsroom/news/2017/7/partnering-with-ilo-in-pakistan
https://www.storaenso.com/-/media/files/documents/sun2022-tyovoimahallinta.pdf
https://www.storaenso.com/en/newsroom/news/2015/11/investing-in-childrens-rights
https://www.storaenso.com/en/sustainability/stakeholder-relations
https://www.storaenso.com/fi-fi/about-stora-enso/stora-enso-locations/sunila-mill
https://www.storaenso.com/fi-fi/newsroom/press-releases/2022/5/stora-enson-yli-800-kesatyontekijaa-tyoskentelee-70-eri-tehtavassa
https://www.storaenso.com/-/media/documents/download-center/documents/suppliers/scoc-2021/stora-enso_supplier-code-of-conduct_2021_v1.ashx
https://www.storaenso.com/en/newsroom/press-releases/2021/10/stora-enso-launches-new-fluff-pulp-grade-with-lower-carbon-footprint
https://www.storaenso.com/en/sustainability/circular-bioeconomy
https://www.storaenso.com/en/newsroom/press-releases/2019/11/stora-enso-and-cepi-member-companies-launch-4evergreen-an-alliance-to-promote-fiber-based-packaging-in-circular-economy
https://www.storaenso.com/en/sustainability/biodiversity/goal-for-net-positive-impact/species
https://yle.fi/news/3-12140859
https://www.storaenso.com/en/sustainability/economic/suppliers
https://www.storaenso.com/fi-fi/newsroom/press-releases/2021/10/stora-enso-on-tehnyt-yrityskohtaisen-tyoehtosopimuksen-paperiliiton-kanssa
https://www.storaenso.com/fi-fi/about-stora-enso/stora-enso-locations/sunila-mill/worksite-information
https://www.storaenso.com/-/media/documents/download-center/certificates/health-and-safety/45001_eng_20220524_qr_biomaterials.pdf
https://www.storaenso.com/fi-fi/newsroom/regulatory-and-investor-releases/2022/4/stora-enson-osavuosikatsaus-tammi-maaliskuulta-2022-erinomainen-tulos-epavakaassa-liiketoimintaymparistossa
https://www.storaenso.com/en/sustainability
https://www.storaenso.com/en/investors/annual-report
https://www.storaenso.com/en/careers/rewards-and-benefits
https://www.storaenso.com/en/sustainability/environmental/forests-plantations-and-land-use
https://mb.cision.com/Public/13589/3285128/81d89b2bc797ed7f.pdf
https://www.storaenso.com/en/bpp/safeguarding-corporate-assets-and-corporate-identity
https://www.storaenso.com/-/media/documents/download-center/documents/suppliers/se_gtc_goods_services_march_2022.pdf
https://www.storaenso.com/en/legal-notice
https://www.storaenso.com/fi-fi/careers/students-and-graduates
https://www.storaenso.com/fi-fi/newsroom/news/2019/8/bio-based-carbon-materials-for-batteries
https://www.storaenso.com/fi-fi/newsroom/regulatory-and-investor-releases/2022/7/stora-enso-ja-northvolt-ryhtyvat-kehittamaan-yhdessa-puupohjaisia-akkuja
https://www.storaenso.com/-/media/documents/download-center/documents/annual-reports/2020/storaenso_sustainability_2020.pdf
https://www.storaenso.com/-/media/documents/download-center/documents/annual-reports/2021/storaenso_annual_report_2021.ashx

Table 8 Example Prompts – Child Labor

Query 1 (Stage 3)	Child labor is work that deprives children of their childhood, their potential, and/or their dignity, and that is harmful to their physical and/or mental development. In its most extreme forms, child labor involves children being enslaved, separated from their families, exposed to serious hazards and illnesses, and/or left to fend for themselves on the streets. Your task is to summarize, based on the documents, whether and to what extent the company works towards the eradication of child labor and pro-actively raise awareness of issues associated with child labor. Your assessment aims to verify if the organization might or is employing children (as defined in the ILO conventions), if the conditions are favorable for the occurrence and/or eradiation of child labor, and the existence and quality of the prevention and mitigating measures taken by the organization. Avoid using jargon language, be concise and clear, and deliver only information that is retrieved in the text. If available, capture concise details and protocols mentioned in the report, including any examples or cases. Do not exceed 300 words in your response. If there is no discussion or mention of the topic, respond "no information" and do not elaborate
Query 2 (Stage 4)	Level − 2: Incidents of child labor, including the worst forms of child labor and children in Hazardous work, have been discovered. Indicators: Absence of positive information while the company is in an area where this situation often occurs according to statistics NGOs reports, voices from local communities, or local schools that indicate child labor Level − 1: Incidents of child labor but no incidents referring to the worst forms of child labor or children in Hazardous work have been discovered within the company or facility, and a corrective action plan with a clear timeline for completion has been developed. Indicators: While the company is in an area where this situation often occurs according to statistics, there is evidence that the company has started to address the situation with a clearly defined timeline presence of organization documents, Interview with directors or human resources officer that indicate no worst forms of child labor, and/or a corrective action plan Level 0: No incidents of child labor are discovered, and the company or facility has a management system in place that enforces the policy prohibiting child labor. Indicators: Records proof of age upon recruitment, including copies of documents such as birth certificates, passports, or religious, medical, or other records. The company is not in an area where child labor is likely, and there is no reason to suspect child labor Level 1: The company or facility has a management system in place to raise awareness of issues associated with child labor within its organization, its subcontractors and its tier 1 suppliers, with the aim to address the root courses of child labor in the region it operates, and takes action as appropriate, like improving the accessibility to schools. Indicators: Evidence that describes the management system of the organization, its subcontractors, and its tier 1 suppliers is effective, for instance, by embedding several of the following examples: Amended job descriptions to include responsibility for children's rights. Staff are trained, empowered, and incentivized to deliver on child rights goals. There is a sensitive grievance mechanism for child rights issues. There is an ombudsperson for children Level 2: The company has made eliminating the use of child labor in its own organization, its entire value chain, including clients, a top priority, and can demonstrate the success of its approach. Indicators: Presence of a child labor project related with detailed informations, monitoring data about pre and post situations in its own organization and along supply chain Your task is to evaluate the given text and determine which level it belongs to. Indicate your judgment in quotation regarding the level for example if the given description falls under Level + 1, indicate "Level + 1" and then after ":" give your chain of thought reasoning for your choice based on the given text and the definitions and performance indicators provided above

Author contributions All authors contributed to the study conception and design. The AI methodology was developed by Arash Hajikhani and Carolyn Cole. The S-LCA methodology was developed in the frame of the ORIENTING project (Zanchi et al. 2023) and tested on a case study by Hanna Pihkola, Essi Paronen, and Eveliina Hylkilä. Data collection and the manual assessment were performed by Eveliina Hylkilä and Essi Paronen. Data preparation and implementation of the AI method were performed by Carolyn Cole. All authors contributed to the comparative analysis. The first draft of the manuscript was written by Carolyn Cole, Arash Hajikhani, Eveliina Hylkilä, and Essi Paronen, and all authors contributed to editing and review of the manuscript. All authors read and approved the final manuscript.

Funding Open Access funding provided by Technical Research Centre of Finland. The work was conducted in the frame of the ORIENTING project that has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 958231.

Data availability The data used and produced in this study can be made available upon request as far as privacy and copyright allow.

Declarations

Conflict of interest The authors declare no competing interests

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Song Z, Hwang GY, Zhang X, Huang S, Park BK (2025) A scientific-article key-insight extraction system based on multi-actor of fine-tuned open-source large language models. Scientific Reports 2025 15:1 15:1–11. https://doi.org/10.1038/s41598-025-85715-7

Abeysinghe B, Circi R (2024) The challenges of evaluating LLM applications: an analysis of automated, human, and LLM-based approaches. CEUR Workshop Proc 3752:4–18

Anagnostopoulos C, Rahman SA, Rizvi SWF, Mamo Y (2025) Leveraging artificial intelligence (AI) for data collection and analysis: the case of and application in corporate social responsibility (CSR) through sport research. International Journal of Sports Marketing and Sponsorship Ahead-of-Print. https://doi.org/10.1108/IJSMS-09-2024-0261/FULL/PDF

Arora C, Grundy J, Abdelrazek M (2024) Advancing requirements engineering through generative AI: assessing the role of LLMs. Generative AI for Effective Software Development 129–148. https://doi.org/10.1007/978-3-031-55642-5_6

Arvidsson R, Hildenbrand J, Baumann H, Islam KMN, Parsmo R (2018) A method for human health impact assessment in social LCA: lessons from three case studies. Int J Life Cycle Assess 23:690–699. https://doi.org/10.1007/S11367-016-1116-7

Arzoumanidis I, Salomone R, Petti L, Mondello G, Raggi A (2017) Is there a simplified LCA tool suitable for the agri-food industry? An assessment of selected tools. J Clean Prod 149:406–425. https://doi.org/10.1016/J.JCLEPRO.2017.02.059

Bai C, Dallasega P, Orzes G, Sarkis J (2020) Industry 4.0 technologies assessment: a sustainability perspective. Int J Prod Econ 229:107776. https://doi.org/10.1016/J.IJPE.2020.107776

Devlin J, Chang MW, Lee K, Toutanova K (2018) BERT: pre-training of deep bidirectional transformers for language understanding. NAACL HLT 2019 - 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies - Proceedings of the Conference 1:4171–4186

Bronzini M, Nicolini C, Lepri B, Passerini A, Staiano J (2024) Glitter or gold? Deriving structured insights from sustainability reports via large language models. EPJ Data Sci 13:41. https://doi.org/10.1140/EPJDS/S13688-024-00481-2

Brown TB, Mann B, Ryder N, Subbiah M, Kaplan J, Dhariwal P, Neelakantan A, Shyam P, Sastry G, Askell A, Agarwal S, Herbert-Voss A, Krueger G, Henighan T, Child R, Ramesh A, Ziegler DM, Wu J, Winter C, Hesse C, Chen M, Sigler E, Litwin M, Gray S, Chess B, Clark J, Berner C, Mccandlish S, Radford A, Sutskever I, Amodei D (2020) Language models are few-shot learners. In: 34th Conference on Neural Information Processing Systems

Sun G, Zhan X, Such J (2024) Building better AI agents: a provocation on the utilisation of persona in LLM-based conversational agents. Proceedings of the 6th Conference on ACM Conversational User Interfaces, CUI 2024. https://doi.org/10.1145/3640794.3665887

Chhipi-Shrestha GK, Hewage K, Sadiq R (2015) "Socializing" sustainability: a critical review on current development status of social life cycle impact assessment method. Clean Technol Environ Policy 17:579–596. https://doi.org/10.1007/S10098-014-0841-5

Cooper J, Noon M, Jones C, Kahn E, Arbuckle P (2013) Big data in life cycle assessment. J Ind Ecol 17:796–799. https://doi.org/10.1111/JIEC.12069

Ziegler ZM, Melas-Kyriazi L, Gehrmann S, Rush AM (2019) Encoder-agnostic adaptation for conditional language generation. arXiv preprint

Ertz M, Kashav S, Zeng T, Sun S (2024) Socially responsible life cycle assessment: organizational activity for the greater good. Social Responsibility Journal 20:647–681. https://doi.org/10.1108/SRJ-11-2022-0481

Polak MP, Morgan D (2024) Extracting accurate materials data from research papers with conversational language models and prompt engineering. Nature Communications 2024 15:1 15:1–11. https://doi.org/10.1038/s41467-024-45914-8

Goedkoop MJ, de Beer IM, Harmens R, Saling P, Morris D, Florea A, Hettinger AL, Indrane D, Visser D, Morao A, Musoke-Flores E, Alvarado C, Rawat I, Schenker U, Head M, Collotta M, Andro T, Viot JF, Whatelet A (2020) Methodology report product social impact assessment 2020. Amersfoort

UNEP (2020) Guidelines for social life cycle assessment of products and organizations 2020. In: Benoît Norris C, Traverso M, Neugebauer S, Ekener E, Schaubroeck T, Russo Garrido S, Berger M, Valdivia S, Lehmann A, Finkbeiner MAG (eds)

Huertas-Valdivia I, Ferrari AM, Settembre-Blundo D, García-Muiña FE (2020) Social life-cycle assessment: a review by bibliometric analysis. Sustainability 2020, Vol 12, Page 6211 12:6211. https://doi.org/10.3390/SU12156211

Jiang S, Hu J, Magee CL, Luo J (2024) Deep learning for technical document classification. IEEE Trans Eng Manag 71:1163–1179. https://doi.org/10.1109/TEM.2022.3152216

ORIENTING D2.3 (2022) LCSA methodology to be implemented in WP4 demonstrations

Lewis P, Perez E, Piktus A, Petroni F, Karpukhin V, Goyal N, Küttler H, Lewis M, Yih W, Rocktäschel T, Riedel S, Kiela D (2020) Retrieval-augmented generation for knowledge-intensive NLP Tasks. Adv Neural Inf Process Syst 33:9459–9474

Min B, Ross H, Sulem E, Veyseh AP Ben, Nguyen TH, Sainz O, Agirre E, Heintz I, Roth D (2021) Recent advances in natural language processing via large pre-trained language models: a survey. ACM Comput Surv 56. https://doi.org/10.1145/3605943

Nascimento DLM, Alencastro V, Quelhas OLG, Caiado RGG, Garza-Reyes JA, Lona LR, Tortorella G (2019) Exploring Industry 4.0 technologies to enable circular economy practices in a manufacturing context: a business model proposal. J Manuf Technol Manag 30:607–627. https://doi.org/10.1108/JMTM-03-2018-0071

Notarnicola B, Sala S, Anton A, McLaren SJ, Saouter E, Sonesson U (2017) The role of life cycle assessment in supporting sustainable agri-food systems: a review of the challenges. J Clean Prod 140:399–409. https://doi.org/10.1016/J.JCLEPRO.2016.06.071

Pihkola H, Hylkilä E, Paronen E, Markkula A, Liirus H (2024) Social-LCA providing new insights and learnings for social responsibility—case pulp and paper industry. The International Journal of Life Cycle Assessment 2024:1–16. https://doi.org/10.1007/S11367-024-02374-Y

Pollok L, Spierling S, Endres HJ, Grote U (2021) Social life cycle assessments: a review on past development, advances and methodological challenges. Sustainability (Switzerland) 13:10286. https://doi.org/10.3390/SU131810286/S1

Raffel C, Shazeer N, Roberts A, Lee K, Narang S, Matena M, Zhou Y, Li W, Liu PJ (2020) Exploring the limits of transfer learning with a unified text-to-text transformer. J Mach Learn Res 21:1–67

Ramos Huarachi DA, Piekarski CM, Puglieri FN, de Francisco AC (2020) Past and future of social life cycle assessment: historical evolution and research trends. J Clean Prod 264:121506. https://doi.org/10.1016/j.jclepro.2020.121506

Rask M, Shimizu K (2024) Beyond the average: exploring the potential and challenges of large language models in social science research. International Conference on Artificial Intelligence, Computer, Data Sciences, and Applications, ACDSA 2024. https://doi.org/10.1109/ACDSA59508.2024.10467341

Gao Y, Xiong Y, Gao X, Jia K, Pan J, Bi Y, Dai Y, Sun J, Guo Q, Wang M, Wang H (2023) Retrieval-augmented generation for large language models: a survey. arXiv preprint

Sassanelli C, Rosa P, Rocca R, Terzi S (2019) Circular economy performance assessment methods: a systematic literature review. J

Clean Prod 229:440–453. https://doi.org/10.1016/J.JCLEPRO.2019.05.019

Shankar S, Zamfirescu-Pereira JD, Hartmann B, Parameswaran AG, Arawjo I (2024) Who validates the validators? Aligning LLM-assisted evaluation of LLM outputs with human preferences. arXiv Preprint

Sica D, Esposito B, Malandrino O, Supino S (2022) The role of digital technologies for the LCA empowerment towards circular economy goals: a scenario analysis for the agri-food system. Int J Life Cycle Assess 1:1–24. https://doi.org/10.1007/S11367-022-02104-2

Tokede O, Traverso M (2020) Implementing the guidelines for social life cycle assessment: past, present, and future. Int J Life Cycle Assess 25:1910–1929. https://doi.org/10.1007/s11367-020-01814-9

Törnberg P (2024) Large language models outperform expert coders and supervised classifiers at annotating political social media messages. Soc Sci Comput Rev. https://doi.org/10.1177/08944393241286471/SUPPL_FILE/SJ-PDF-1-SSC-10.1177_08944393241286471.PDF

Tragnone BM, Arzoumanidis I, D'Eusanio M, Petti L (2023) Contribution of the product social impact life cycle assessment (PSILCA) database in assessing the risks and opportunities of a jar of honey production. Int J Life Cycle Assess 28:1054–1071. https://doi.org/10.1007/S11367-023-02171-Z

UNEP (2021) Methodological sheets for subcategories in social life cycle assessment (S-LCA) 2021. In: Traverso M, Valdivia S, Luthin A, Roche L, Arcese G, Neugebauer S, Petti L, D'Eusanio M, Tragnone BM, Mankaa R, Hanafi J, Benoît Norris C, Zamagni A (eds). United Nations Environment Programme (UNEP)

Wang Y, Han JH, Beynon-Davies P (2019) Understanding blockchain technology for future supply chains: a systematic literature review and research agenda. Supply Chain Manag 24:62–84. https://doi.org/10.1108/SCM-03-2018-0148

Wang Z, Nayfeh T, Tetzlaff J, O'Blenis P, Murad MH (2020) Error rates of human reviewers during abstract screening in systematic reviews. PLoS ONE 15:e0227742. https://doi.org/10.1371/JOURNAL.PONE.0227742

Zhang A, Zhong RY, Farooque M, Kang K, Venkatesh VG (2020) Blockchain-based life cycle assessment: an implementation framework and system architecture. Resour Conserv Recycl 152:104512. https://doi.org/10.1016/J.RESCONREC.2019.104512

Artificial Analysis (2025) LLM performance leaderboard - a hugging face space by artificialanalysis. https://huggingface.co/spaces/ArtificialAnalysis/LLM-Performance-Leaderboard. Accessed 22 Apr 2025

Chalkidis I, Fergadiotis M, Malakasiotis P, Aletras N, Androutsopoulos I (2020) LEGAL-BERT: the Muppets straight out of law school. Findings of the Association for Computational Linguistics Findings of ACL: EMNLP 2020 2898–2904. https://doi.org/10.18653/v1/2020.findings-emnlp.261

Fischer T, Biemann C (2024) Exploring large language models for qualitative data analysis. Proceedings of the 4th International Conference on Natural Language Processing for Digital Humanities 423–437. https://doi.org/10.18653/V1/2024.NLP4DH-1.41

Liu F, Vulić I, Korhonen A, Collier N (2021) Learning domain-specialised representations for cross-lingual biomedical entity linking. ACL-IJCNLP 2021 - 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing, Proceedings of the Conference 2:565–574. https://doi.org/10.18653/v1/2021.acl-short.72

Pokhrel S, Ganesan S, Akther T, Karunarathne L (2024) Building customized chatbots for document summarization and question answering using large language models using a framework with OpenAI, LangChain, and Streamlit. Journal of Information Technology and Digital World 6:70–86. https://doi.org/10.36548/JITDW.2024.1.006

Topsakal O, Akinci TC (2023) Creating large language model applications utilizing LangChain: a primer on developing LLM apps fast. International Conference on Applied Engineering and Natural Sciences 1:1050–1056. https://doi.org/10.59287/ICAENS.1127

TruEra (2024a) TruLens RAG Triad. https://www.trulens.org/trulens_eval/getting_started/core_concepts/rag_triad/. Accessed 12 Apr 2024

TruEra (2024b) GitHub - truera/trulens: evaluation and tracking for LLM experiments. https://github.com/truera/trulens

Wang A, Singh A, Michael J, Hill F, Levy O, Bowman SR (2018) GLUE: a multi-task benchmark and analysis platform for natural language understanding. EMNLP 2018 - 2018 EMNLP Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP, Proceedings of the 1st Workshop 353–355. https://doi.org/10.18653/v1/w18-5446

Yoo KM, Park D, Kang J, Lee SW, Park W (2021) GPT3Mix: leveraging large-scale language models for text augmentation. Findings of the Association for Computational Linguistics, Findings of ACL: EMNLP 2021 2225–2239. https://doi.org/10.18653/V1/2021.FINDINGS-EMNLP.192

Zanchi L, Zamagni A, Pillay S, Mangana D, Pihkola H, Paronen E, Hylkilä E, Sonderegger T, Martinez EC, Saling P, Isasa M, Ugarte R (2023) Specification of social indicators for LCSA. EU Horizon 2020 project ORIENTING (GA No 958231). https://ec.europa.eu/research/participants/documents/downloadPublic?documentIds=080166e507e98507&appId=PPGMS

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.